Bounding Negative Transfer for Federated Multi-Source Cross-Domain Recommendation

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***Abstract*—Federated Cross-Domain Recommendation (CDR), where CDR is integrated with federated learning, is attracting much interest. Existing literature focuses on one or two source domains. We propose to generalize to multi-source CDR, to harness the growing availability of data from different domains. To address the new challenges in privacy protection and negative knowledge transfer brought about by multiple source domains, we propose a model named FEDMCDR with a *knowledge preparation module* and a *positive knowledge activation module*. The knowledge preparation module employs differential privacy-based knowledge extraction and an adaptive mapping function to ensure reliable knowledge transfer. The positive activation module incorporates a reinforcement learning-based knowledge filter that eliminates irrelevant feature dimensions. We further provide a theoretical analysis that bounds the impact of negative transfer under our framework. Extensive experiments over seven domains of the Amazon dataset and three domains of the Douban dataset with various intensities of negative knowledge demonstrate the consistent superiority of FEDMCDR. Compared with the strongest baseline FedGCDR, FEDMCDR achieves a relative improvement of up to 42.0% in HR@10 and 57.5% in NDCG@10.**

***Index Terms*—Cross-domain recommendation, federated learning, graph neural networks, transfer learning.**

I. Introduction

Cross-Domain Recommendation (CDR) transfers knowledge from known *source domains* to enhance recommendation accuracy for a *target domain* [1]. Motivated by growing public concerns over privacy and data protection regulations, e.g., the General Data Protection Regulation of the European Union, recent studies incorporate Federated Learning (FL) into CDR and propose federated CDR models [2]–[7].

A few federated CDR studies (**Class I**) [5], [8] focus on organization-level privacy, i.e., *vertical FL*, where each domain server (e.g., corresponding to an organization) contains features of the same users in different dimensions (domains, cf. Fig. 1 (a)). These models maintain centralized storage for each domain and apply privacy-preserving techniques to facilitate encrypted knowledge transfer between domains. Other studies (**Class II**) [2], [3], [6], [7] consider user-level privacy, i.e., *horizontal FL*, where sensitive user data are stored locally (i.e., decentralized on user device, cf. Fig. 1 (b)). Knowledge transfer happens locally in plain text at the user side, while the server of each domain trains a recommendation model with all associated users through FL.

These studies focus on knowledge transfer from *one or two* data-rich source domains to a data-sparse target domain, e.g., to address the cold-start problem [1]. As user data from more domains become available, e.g., from a variety of apps providing different services, a new opportunity arises: to transfer knowledge from *multiple* source domains to strengthen the recommendation quality of the target domain (Fig. 1 (c)). Such an opportunity, however, has been under-explored.

To fill this gap, we propose to study a more general problem, *federated multi-source CDR*, with *two or more source domains*. We aim to learn more complete user preference profiles from multiple source domains to achieve a more accurate recommendation for the target domain, with privacy constraints. Our study aims to create new opportunities for apps to share user insights with each other to improve service quality (*utility*) without compromising user data privacy.

The additional source domains bring two new challenges.

(1) Source-domain side: How to enable trustworthy knowledge extraction under multi-source privacy–utility trade-offs. In federated multi-source CDR, source domains must strike a balance between privacy-preservation and effective feature sharing to prevent from disrupting the target domain's model performance. Existing federated CDR studies face a significant dilemma: Class I models transfer encrypted embeddings between organizations but struggle to model complex cross-domain dependencies [5], [8], while Class II models manage embeddings locally in plain text, which is vulnerable to privacy leaks across domains [2], [3]. As shown in Fig. 1 (c), these frameworks become ineffective in multi-source scenarios because privacy-induced distortion, coupled with heterogeneous source quality (e.g., varying data volumes, biases, and training errors), makes the transferred knowledge inconsistent and error-prone. If the privacy-utility trade-off is managed poorly, more source domains could lead to more severe negative knowledge transfer that poisons the target domain model, as illustrated in Fig. 1 (d).

(2) Target-domain side: How to achieve selective knowledge activation under multi-source heterogeneity and conflicts. In federated multi-source CDR, even when source domains provide nominally trustworthy knowledge, the target domain remains susceptible to redundant, irrelevant, or negative transfer due to complex domain discrepancies and cross-source conflicts. Existing federated CDR methods lack robust knowledge selection mechanisms [2] or rely on rigid attention

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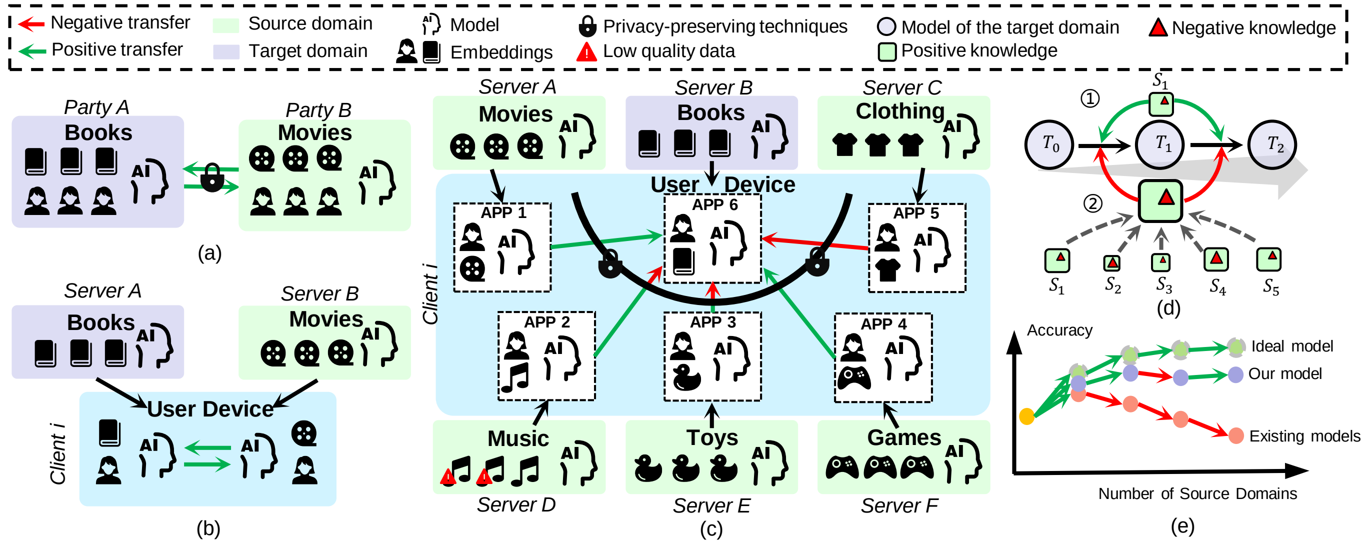

Fig. 1. Federated CDR settings. (a) Class I: organization-level privacy (vertical FL). (b) Class II: user-level privacy (horizontal FL). (c) *Federated multi-source CDR* (ours): a target domain (e.g., `Books`) leverages knowledge from multiple source domains (e.g., `Movies`, `Clothing`, `Music`, `Toys`, and `Games`). Each domain can be regarded as a different app on a user device (e.g., a smartphone). (d) Accumulated negative source-domain knowledge. Negative transfer accumulates more easily in the multi-source setting than dual-domain settings, without balancing privacy and utility. (e) Performance degradation with more source domains. Target-domain performance can degrade as the number of source domains increases if negative knowledge is not properly filtered.

schemes [3], [9] that work for dual-domain settings but fail when there are more source domains. These methods fail to adaptively activate knowledge from each different source domain. Without an effective selective activation strategy, the target model may struggle to derive useful knowledge from multiple source domains, resulting in performance declines with more source domains [10], as shown in Fig. 1 (e).

To address these challenges, we propose FEDMCDR, a federated multi-source CDR framework based on Graph Attention Networks (GAT). We adopt GAT as the base model, as it is a popular backbone in recent recommender systems [11], has been shown to accurately learn user and item representations, and provides a means to mitigate negative transfer [10], [12].

FEDMCDR takes a *horizontal-vertical-horizontal* federated transfer learning pipeline. The first horizontal stage trains a source model (a GAT) for each source domain under horizontal FL setting, where each individual user (with its device) is a client. The last horizontal stage trains the target model (another GAT), also with horizontal FL where each user is a client.

The middle, vertical stage is our core novel design, supported by two key modules: a *knowledge preparation* module and a *positive knowledge activation* module. The source-side challenge is addressed by the privacy-preserving knowledge preparation module, while the target-side challenge is tackled by the positive knowledge activation module.

The knowledge preparation module extracts user embeddings from the propagation layers of each source-domain GAT and applies Differential Privacy (DP) [13] to these embeddings to prevent privacy leaks. To mitigate the negative effect of DP noise on target-domain model training while preserving the utility of source-domain knowledge, this module then adopts an adaptive mapping function that projects the encrypted knowledge vectors (i.e., user embeddings) into a feature space suitable for target-domain model training. The resulting user embeddings have a better privacy–utility trade-off and are transferred to the target domain.

The positive knowledge activation module then suppresses negative transfer by using a pre-trained policy agent to filter irrelevant or potentially harmful embedding dimensions. Based on the filtered knowledge vectors, this module expands the local graph of the target domain for each client by introducing virtual user nodes and edges, which enables domain-attention calculation and activates positive cross-domain transfer during target-domain model training on the expanded graph.

We quantify the negative knowledge transferred from the source domains and the impact on the target-domain model through deriving upper bounds on the output deviation of the target-domain model. To the best of our knowledge, this work presents the first quantitative theoretical analysis of negative transfer in federated CDR.

Overall, this paper makes the following contributions.

C1. We propose a more general federated CDR problem, federated multi-source CDR with two or more source domains.

C2. We propose a framework named FEDMCDR for the problem, with a horizontal-vertical-horizontal pipeline to address negative transfer issues while ensuring user-data privacy.

C3. We propose two core modules for FEDMCDR. The knowledge preparation module protects user privacy through differential privacy and enables training-step-varying knowledge transfer through an adaptive mapping function. The positive knowledge activation module filters negative knowledge and expands the local graph on each user's side with transferred knowledge to enhance target-domain model training.

C4. We quantify the negative knowledge from the source domains and derived upper bounds of its impact on the predictions of the target-domain model.

C5. We conduct extensive experiments on seven domains of the Amazon datasets and three domains of the Douban dataset. The results demonstrate the effectiveness of FEDMCDR. It outperforms the strongest baseline FedGCDR [10] consistently, achieving performance improvements up to 42.0% in HR@10 and 57.5% in NDCG@10. *Source code of* FEDM-

CDR *is included in supplementary materials and will be released upon paper acceptance.*

This paper is an extension of our previous conference paper [10]. The conference version presented the federated multi-source CDR problem (named BS-CDR there, C1) and a framework for the problem with the horizontal-vertical-horizontal pipeline (named FedGCDR there, C2). We also showed that the framework retains differential privacy for user embeddings from the source domains.

In this extension, we strengthen both the knowledge preparation and the positive knowledge activation modules with new designs (C3). For knowledge preparation, we propose an adaptive mapping function (**Section IV-B**). This design enables dynamical adjustment of the scale of knowledge to be transferred according to the training process of the target domain. For positive knowledge activation, we propose a policy agent to filter negative knowledge at a more granular dimension-level, and remove the fine-tuning stage of the target-domain model (**Section IV-C**). We further quantify the negative knowledge and bound its impact on the predictions of the target-domain model (C4, **Section V**). We conduct experiments with the new modules and the FEDMCDR framework showing their effectiveness (C5, **Section VII**). We also rewrote the paper for better readability.

II. Related work

Federated cross-domain recommendation. A few studies consider each domain to be with a different organization and focus on protecting organization-level privacy with encrypted knowledge transfer. For example, Chen *et al.* [5] utilize orthogonal mapping matrices with local differential privacy (LDP) to protect privacy for inter-organization transfer, while Liu *et al.* [8] share only insensitive distribution statistics derived from Graph Convolutional Networks (GCNs) to minimize leakage. These studies rely on centralized data storage within each domain, ignoring the associated privacy concerns.

More recent studies focus on protecting user privacy by treating users as individual clients, adopting decentralized storage. Liu *et al.* [2] first define the Edge-CDR problem, modifying the target of privacy protection from organizations to users. They propose a VAE-based federated framework to extract cross-domain knowledge and safeguard user privacy. Subsequent works focus on addressing unique challenges in the decentralized, edge-device based and federated scenario. Wu *et al.* [3] proposed an MLP-based personalized function to learn local and personal user vectors and solve the heterogeneity challenge in data or user preference, under federated setting. Tian *et al.* [6] design two contrastive loss which reduce deviations between the global model and the local model, as well as the deviations between the current model and the historical model, to tackle the issue of model drift.

These solutions struggle with balancing privacy with utility for *federated multi-source CDR* settings. Existing solutions that transfer user embeddings in plain text [2], [3] prioritize utility but leave users vulnerable to inference attacks. In contrast, solutions that incorporate standard privacy techniques (e.g., LDP) [6] often inadvertently disrupt the complex feature correlations required for multi-source alignment.

To fill the gaps, we propose a model that decentralizes user data storage and adopts DP-based knowledge extraction, with an adaptive mechanism to resolve domain conflicts.

Negative transfer in cross-domain recommendation. Recent studies find that directly transferring source-domain knowledge could be harmful; complex inter-domain discrepancies could negatively impact target model accuracy, a phenomenon known as *negative transfer* [10], [12].

Existing solutions primarily follow two paradigms to mitigate this issue. The first paradigm focuses on extracting domain-invariant (or shared) features to bridge domain gaps [14]–[16]. While effective in dual-domain settings (i.e., single source domain), it remains difficult to identify features shared across *all* domains in multi-source domain settings [12], as the intersection of shared information shrinks significantly with the increase of heterogeneous sources.

The second paradigm introduces attention mechanisms or manual re-weighting to promote positive transfer. Several studies [9], [17] compute domain attention by measuring the similarity of predefined features between the source and target domains. Others, such as FedCDR [3] and EMCDR [18], employ hyper-parameters to control the transfer intensity. These solutions face scalability issues in multi-source settings. Predefined features are difficult to accommodate the diverse and unstructured nature of multiple domains [12], while manually tuning attention-related hyper-parameters becomes intractable due to the combinatorial complexity.

These issues limit existing solutions to few source domains. They lack a scalable mechanism to handle the cumulative negative transfer in multi-source settings. We propose a model that adapts to multi-source settings to address these issues.

III. Preliminaries

We consider M ($M \geq 2$) source domains, denoted as $\mathcal{D}^{S_1}, \mathcal{D}^{S_2}, \ldots, \mathcal{D}^{S_M}$, and a target domain $\mathcal{D}^T$, i.e., a total of $M+1 (\geq 3)$ domains. Each domain corresponds to *a domain server* that performs intra-domain model training. There is a global user set U shared by all domains. Each user $u \in U$ is considered an individual client who is connected to all domain servers, i.e., their local device contains domain models from all source and target domains.

For each domain i ($i \in [1, M+1]$, the first M domains are the source, and the last domain is the target; same below), there is a set of items V^i. We use $\mathbf{R}^i$ which is a $|U| \times |V^i|$ binary matrix to denote the observed *interaction matrix* between the users in set U and the items in set V^i. Each entry $r^i_{jk} \in \mathbf{R}^i$ is in $\{0, 1\}$, where 1 indicates an interaction between user j and item k, and 0 otherwise.

Problem statement. Given a user set U, item sets $V^1, V^2, \ldots, V^{M+1}$, and the corresponding interaction matrices $\mathbf{R}^1, \mathbf{R}^2, \ldots, \mathbf{R}^{M+1}$, our aim is to predict for $\mathbf{R}^{M+1}$ (i.e., the interaction matrix of the target domain) the zero-valued entries r^{M+1}_{jk} that are most likely to turn to 1. Intuitively, we aim to find target-domain items in V^{M+1} that are most likely to attract interactions with the users in U, for recommendation.

As discussed in Section I and Fig 1 (c), our setting aligns with real-world scenarios where domains correspond to apps

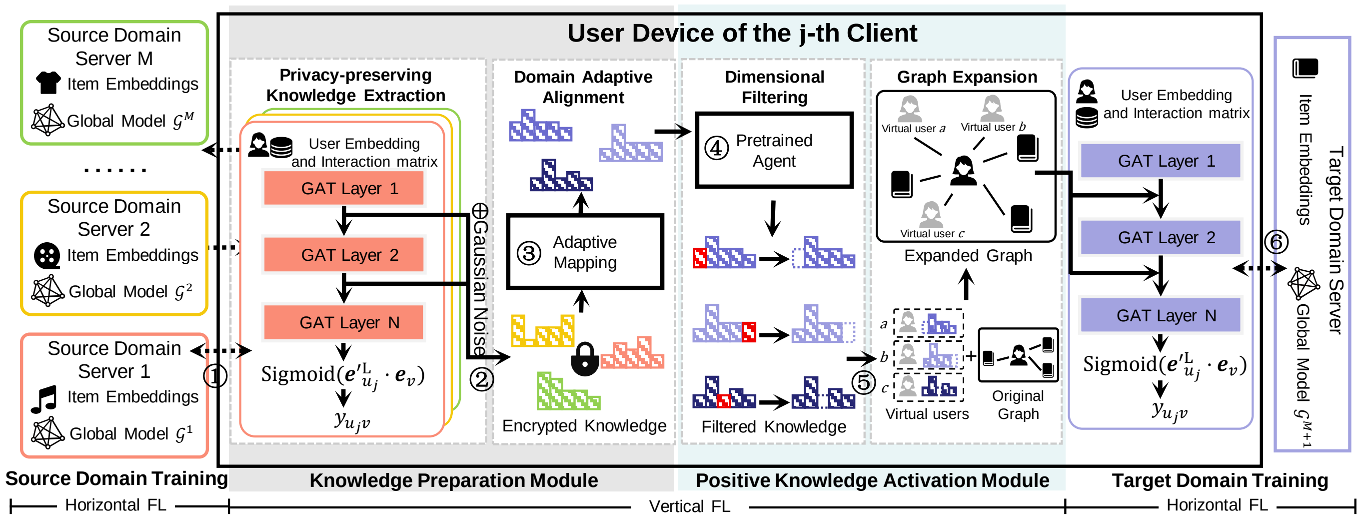

Fig. 2. Overview of our model FEDMCDR. The model consists of four stages and follows a horizontal-vertical-horizontal FL pipeline: (I) **Source Domain Training** (Horizontal FL): ① Each source domain train its GAT-based federated model. (II) **Knowledge Preparation** (Vertical FL): ② User embeddings of the source domains are extracted from GAT layers and perturbed with Gaussian noise. ③ The perturbed and encrypted embeddings are transmitted to an adaptive mapping function, mapping and rescaling these embeddings to facilitate an adaptive transfer. (III) **Positive Knowledge Activation** (Vertical FL): ④ A policy agent filters the irrelevant dimensions of the transferred knowledge, to mitigate "negative transfer". ⑤ The local user-item interaction graph of each user is expanded with the filtered source-domain knowledge. (IV) **Target Domain Training** (Horizontal FL): ⑥ The target domain then trains its GAT-based federated model and makes recommendations accordingly.

on users' mobile devices, and domain servers represent the respective service providers. Popular apps are used by many users. Their shared set of users is modeled by the user set U. Our study opens new opportunities for such apps to share user insights, for enhancing service quality and user experience without compromising user or business data privacy.

Graph attention networks in recommender systems. For recommender systems, a GAT [19] consists of $L \in \mathbb{Z}^+$ message propagation layers and is typically trained on the bipartite graph that represents the interaction matrix between users and items [20]. At the l-th layer, embedding $\mathbf{e}_u^l$ of each user node $u \in U$ is transformed using a shared linear projection (the same for each item node $v \in V$ which is omitted for conciseness; same below), where $\mathbf{W}^l \in \mathbb{R}^{K \times K'}$ – K is the dimensionality of $\mathbf{e}_u^l$, and K' is that of $\mathbf{e}'^l_u$:

$$\mathbf{z}_u^l = \mathbf{W}^l \mathbf{e}_u^l, \tag{1}$$

The attention coefficient for node u towards node v is:

$$\alpha_{uv}^l = \frac{\exp\big(\mathrm{LeakyReLU}(\mathbf{a} \cdot (\mathbf{z}_u^l || \mathbf{z}_v^l))\big)}{\sum_{v' \in N(u) \bigcup u} \exp\big(\mathrm{LeakyReLU}(\mathbf{a} \cdot (\mathbf{z}_u'^l || \mathbf{z}_{v'}'^l))\big)}, \tag{2}$$

where $\mathbf{a}$ is a learnable attention vector, '$||$' is the concatenation operation, and $N(u)$ is the neighbor set of u. The node embeddings for the next layer are computed by aggregating the weighted, transformed embeddings of the neighboring nodes:

$$\mathbf{e}'^l_u = \alpha_{uu}^l \mathbf{z}_u^l + \sum_{v \in N(u)} \alpha_{uv}^l \mathbf{z}_v^l, \quad \mathbf{e}'^l_v = \sum_{u \in N(v)} \alpha_{vu}^l \mathbf{z}_u^l + \alpha_{vv}^l \mathbf{z}_v^l. \tag{3}$$

Embedding $\mathbf{e}'^l$ is fed into the next GAT layer as $\mathbf{e}^{l+1}$, to facilitate the following message propagation in layer $l+1$.

IV. METHODOLOGY

As shown in Fig. 2, our framework FEDMCDR takes a horizontal-vertical-horizontal (HVH) FL pipeline. The vertical stage has two core modules, a knowledge preparation module and a positive knowledge activation module. We detail the pipeline and the two modules in this section.

A. Horizontal-Vertical-Horizontal FL pipeline

Our HVH FL pipeline consists of three main stages that alternate between horizontal and vertical FL.

The first stage applies horizontal FL for source-domain (recommendation) model training. Each source domain server individually communicates with its domain clients (individual users) to collaboratively train a GAT model $\mathcal{G}^i$. The private user-item interaction matrix $\mathbf{R}^i$ is distributed and stored locally within each client (hence horizontal FL). The clients only exchange model parameters and gradients with the servers.

The next stage applies vertical FL. It extracts user preference embeddings from each GAT $\mathcal{G}^i$ of the source domains, applies DP on them for privacy protection, and maps the encrypted dimensions while filtering unhelpful dimensions to mitigate negative transfer. The resulting user preference embeddings are used as virtual user nodes to expand the user-item graph of the target domain. This stage is *fully computed on each client device* (e.g., a smart phone).

The final stage applies horizontal FL again, to train the target-domain model $\mathcal{G}^{M+1}$ in a distributed manner as above.

GAT backbone model. We adopt GAT as the backbone model for each domain. For domain $i \in [1, M+1]$, we assign each user $u \in U$ and item $v \in V^i$ a random ID embedding of size d (a hyper-parameter) which serves as the input of the first layer of GAT, denoted as $\mathbf{e}_u^0$ and $\mathbf{e}_v^0$, respectively. The user and item embedding matrices of domain i are denoted as $\mathbf{E}_u \in \mathbb{R}^{|U| \times d}$ and $\mathbf{E}_v \in \mathbb{R}^{|V| \times d}$, respectively. For the structure of GAT $\mathcal{G}_i$ of domain i, inspired by LightGCN [11], we skip feature transformation (Eq. 1) and nonlinear activation

(LeakyReLU in Eq. 2) for efficiency. The attention score α for user u and item v is calculated as follows:

$$\alpha_{uv}^{l} = \frac{\exp(\mathbf{a}\cdot(\mathbf{e}_u^l||\mathbf{e}_v^l))}{\sum_{v'\in N(u)\cup u}\exp(\mathbf{a}\cdot(\mathbf{e}_u^l||\mathbf{e}_{v'}^l))}. \tag{4}$$

In our horizontal FL, each client's local user-item interaction graph includes one user, multiple items, and edges between user and items. The output state of users and items is:

$$\mathbf{e'}_u^l = \alpha_{uu}\mathbf{e}_u^l + \sum_{v\in N(u)}\alpha_{uv}\mathbf{e}_v^l, \quad \mathbf{e'}_v^l = \alpha_{vu}\mathbf{e}_u^l + \alpha_{vv}\mathbf{e}_v^l. \tag{5}$$

The training of $\mathcal{G}_i$ follows the horizontal FL paradigm [21], where only the model's parameters and gradients are exchanged between the domain server and each client. For the source domains, we use the binary cross entropy loss. We will detail the loss function of the target domain in Section IV-D.

The GAT model we adopted is the same as that in FedGCDR [10]. Hence, we inherit its training and privacy guarantee of federated graph learning for a single GAT. These are not our focus and will not be detailed further. Other GAT-based models (e.g, [4], [22]) may be applied as well.

B. Knowledge Preparation Module

After source domain training, we obtain M source models $\mathcal{G}^1, \mathcal{G}^2, \ldots, \mathcal{G}^M$. Each client (a user $u_j \in U$) has a copy of these models (distributed form server during source domain training) and corresponding embeddings on the local device. The knowledge preparation module prepares the knowledge (i.e., user embeddings) from these models to be transferred to the target domain, with source domain privacy protection.

From source domain $\mathcal{D}^{S_i}$, we extract an embedding matrix $\mathbf{X}^{S_i,j} \in \mathbb{R}^{(L-1)\times d}$ for user $u_j \in U$ in the local device, which is formed by the output embeddings of u_j of the first $L-1$ layers of $\mathcal{G}^i$. Row l (counting from 0) of $\mathbf{X}^{S_i,j}$, denoted as $\mathbf{x}_l^{S_i,j}$, is the output embedding from layer l of $\mathcal{G}^i$: $\mathbf{e'}_{u_j}^l$ of $\mathcal{G}^i$.

We transfer $\mathbf{X}^{S_i,j}$ from each source domain i to the target domain for user preference modeling. A direct transfer, i.e., using $\mathbf{X}^{S_i,j}$ to train the target-domain model $\mathcal{G}^{M+1}$, however, causes two problems: (i) It breaks data privacy. An app (i.e., a domain) would not share its model data directly. (ii) Due to domain discrepancy, the transferred $\mathbf{X}^{S_i,j}$ may interfere with the training of the target domain. We address these issues next.

a) Privacy-preserving Knowledge Extraction: An attacker can infer a user's private interaction information based on the transferred raw embeddings [23]. To address this issue, we apply DP [13], [24] to each row of matrix $\mathbf{X}^{S_i,j}$ with the Gaussian mechanism to protect privacy.

Definition 1 (Gaussian Mechanism): Given a function $f: X \to \mathbb{R}^d$ over a dataset X, the Gaussian mechanism f_G is a function defined as:

$$f_G(x, f(\cdot), \epsilon) = f(x) + \mathbf{r}, \tag{6}$$

where $x \in X$, ϵ is the privacy parameter, and $\mathbf{r} \in \mathbb{R}^d$ is the Gaussian noise vector – the value of each dimension is sampled from $\mathcal{N}(0,\sigma)$, where $\sigma = \frac{\Delta_2 f}{\epsilon}\sqrt{2\ln(\frac{1.25}{\delta})}$, with $\Delta_2 f$ being the L_2-sensitivity, and δ the failure probability.

We apply the Gaussian mechanism on each row of $\mathbf{X}^{S_i,j}$, treating $\mathcal{G}^i$ as $f(\cdot)$ and $\mathbf{x}_l^{S_i,j}$ as $f(x)$ in Eq. 6. The resulting vector and the corresponding matrix are denoted as $\tilde{\mathbf{x}}_l^{S_i,j}$ and $\tilde{\mathbf{X}}^{S_i,j}$, respectively. This design ensures that the target domain cannot trivially infer the source-domain user-item interaction relationships $\mathbf{R}^i$ from the *transfer matrix* $\hat{\mathbf{X}}^{S_i,j}$.

Theorem 1: The Gaussian mechanism defined in Definition 1 preserves (ϵ,δ)-DP [13]. By adding Gaussian noise to the source-domain embeddings $\mathbf{X}^{S_i,j}$ to create a transfer matrix $\hat{\mathbf{X}}^{S_i,j}$, an ideal reconstruction of $\mathbf{X}^{S_i,j}$ is prevented.

Proof: This theorem comes naturally from applying DP. The Gaussian noise prevents the target-domain app from deriving $\mathbf{X}^{S_i,j}$ precisely and accessing the user-item interaction matrix $\mathbf{R}^i$. Please refer to Appendix A of our conference version [10] for a full proof. ■

FEDMCDR adopts the same DP mechanism as that in FedGCDR [10]. In this extension, we focus on negative transfer in multi-domain recommendation, instead of developing new privacy-preserving mechanisms.

By integrating decentralized data storage (horizontal FL for source- and target-domain model training) with encrypted knowledge transfer between domains, FEDMCDR addresses two privacy concerns: (i) user privacy leaks inherent to centralized data storage, and (ii) cross-domain privacy leaks, which are an important roadblock in cross-domain data sharing.

b) Domain Adaptive Alignment: The discrepancy between the source and target domains and their respective latent user embedding spaces is an important source of negative transfer. In our conference version [10], we propose to use *multilayer perceptrons* (MLPs) to map user embeddings from each source domain to the target domain:

$$f_M^i(\tilde{\mathbf{x}}_l^{S_i,j}) = f_\sigma(\mathbf{W}^i \cdot \tilde{\mathbf{x}}_l^{S_i,j} + \mathbf{b}), \tag{7}$$

where $\mathbf{W}^i \in \mathbb{R}^{d\times d}$ is a learnable matrix, and $\mathbf{b}^i \in \mathbb{R}^d$ is a learnable bias. To avoid confusion with the standard deviation σ of Gaussian distribution, we denote the activation function by f_σ. We use the MSE loss between $f_M^i(\hat{\mathbf{x}}_l^{S_i,j})$ and target user embeddings $\mathbf{e}_{u_j}^l$ to promote alignment.

We observe two issues with this design: (i) At the early stage of target domain training, randomly initialized MLPs are unable to provide accurate mapping. The noisily mapped knowledge cannot be filtered by the GAT of the target domain, which has also been randomly initialized. These cause the target-domain model to bias towards the source-domain distribution [25]. With more source domains, the chance of training a less accurate MLP is higher [12]. (ii) The learned knowledge is static when it is transferred to the target domain. This static input knowledge, when integrated with the MSE loss, leads the MLPs to map the knowledge vectors ($f_M^i(\tilde{\mathbf{x}}_l^{S_i,j})$) to be similar to the user embeddings of the target domain, thereby losing the source-domain information in $\tilde{\mathbf{x}}_l^{S_i,j}$. This is shown in our ablation study in Section VII-E1.

To address these issues, we propose an *adaptive mapping function*, denoted as $f_A^i(\cdot)$ for domain i:

$$f_A^i(\hat{\mathbf{x}}_l^{S_i,j}) = k^i \cdot \tanh(\mathbf{W}^i \cdot \hat{\mathbf{x}}_l^{S_i,j} + \mathbf{b}^i), \tag{8}$$

where $\tanh(\cdot)$ is the hyperbolic tangent activation function, $\mathbf{W}^i \in \mathbb{R}^{d\times d}$ is a learnable weight matrix, $\mathbf{b}^i \in \mathbb{R}^d$ is a learnable bias vector, and k^i is a learnable coefficient.

The $\tanh(\cdot)$ activation function normalizes the embedding values into the range of $[-1, 1]$ for each dimension. The value of k^i is initialized to be the same as the initialization range of the target-domain embeddings (e.g., when the embeddings of target domain are sampled from uniform distribution $\mathcal{U}(-1, 1)$, k^i is 1), and it will be optimized during target-domain training, reflecting the change of *acceptance capacity* of the target domain. Together, they scale the mapped knowledge to be inline with the magnitude of the user embeddings of the target domain, such that the transferred knowledge is large enough to impact the target-domain embeddings, while it is not too large and dominate the target-domain embeddings. We denote the mapped vector and the matrix as $\mathbf{h}_l^{S_i,j}$ and $\mathbf{H}^{S_i,j}$, respectively.

Further, we drop the MSE loss and treat the adaptive mapping function as a proxy task to better leverage the source-domain knowledge in $\tilde{\mathbf{X}}^{S_i,j}$. At the target-domain training stage, we transfer $\mathbf{H}^{S_i,j}$ to improve the robustness of the target-domain model, enhancing its ability to filter out negative knowledge. At the test stage, we transfer $\tilde{\mathbf{X}}^{S_i,j}$ directly to leverage the original knowledge while avoiding potentially harmful cross-domain mapping (it is hard to ensure the accuracy of every mapping function in multi-source CDR [12]). We conduct relevant ablation study in Section VII-E.

C. Positive Knowledge Activation Module

After knowledge preparation, the source-domain knowledge is transferred to the target domain $\mathcal{D}^T$. Each user u_j (e.g., the target-domain app in u_j's mobile) receives a list of knowledge matrices from the source domains: $\mathbf{H}^{S_1,j}, \mathbf{H}^{S_2,j}, \ldots, \mathbf{H}^{S_M,j}$. Each row of the matrices, $\mathbf{h}_l^{S_i,j}$, represents $f_A^i(\tilde{\mathbf{x}}_l^{S_i,j})$. Our positive activation module aims to leverage the transferred knowledge and further mitigate potential negative transfer.

Embedding $\mathbf{h}_l^{S_i,j}$ is fused into user embedding $\mathbf{e}_{u_j}^l$ of layer l of the GAT of $\mathcal{D}^T$. This way, we exploit the user features from the source domains in each different abstract level. A simple approach is to fuse $\mathbf{h}_l^{S_i,j}$ with $\mathbf{e}_{u_j}^l$ by their average. This, however, may introduce irrelevant or harmful information, due to domain discrepancy or noisy knowledge mapping.

a) Dimensional Filtering: We introduce a pre-trained policy agent to filter the harmful dimensions, and we present its training strategy in three main steps below.

Preparation. The training of the policy agent is based on the target domain's GAT, embeddings, and interaction matrix. First, the target domain uses its own domain interaction data to train a GAT, which is carried out in parallel to the source domain training with the same horizontal FL. Subsequently, we train the policy agent, freezing the target-domain GAT and embeddings, and assigning the policy agent to each layer of the GAT. For the l-th layer, embeddings of u_j (i.e., $\mathbf{e}_{u_j}^l$) and its neighbors ($N(u)$) are fed into the policy agent. For a neighbor $n \in N(u_j)$, the agent evaluates whether the i-th dimension of its embedding ($\mathbf{e}_n^l$) is irrelevant and should be zeroed. To achieve this, we first generate a candidate embedding ${\mathbf{e}'}_n^l$, zeroing only dimension i, and define the following features (for simplicity, we omit the superscript l and use $\mathbf{e}^{(i)}$ to represent the i-th dimension of embedding $\mathbf{e}$):

f_1 : The absolute difference between $\mathbf{e}_{u_j}^{(i)}$ and $\mathbf{e}_n^{(i)}$: $|\mathbf{e}_{u_j}^{(i)} - \mathbf{e}_n^{(i)}|$.

f_2 : The absolute difference between $\mathbf{e}_n^{(i)}$ and the mean value of the i-th dimension over all neighbors of u_j: $|\mathbf{e}_n^{(i)} - \frac{1}{|N(u_j)|}\sum_{n' \in N(u_j)} \mathbf{e}_{n'}^{(i)}|$.

f_3 : The absolute change in the absolute difference between the mean values of vectors $\mathbf{e}_{u_j}$ and $\mathbf{e}_n$ before and after zeroing dimension i: $\big||\operatorname{mean}(\mathbf{e}_{u_j}) - \operatorname{mean}(\mathbf{e}_n)| - |\operatorname{mean}(\mathbf{e}_{u_j}) - \operatorname{mean}({\mathbf{e}'}_n)|\big|$.

f_4 : The absolute change in the absolute difference between the standard deviations of vectors $\mathbf{e}_{u_j}$ and $\mathbf{e}_n$ before and after zeroing dimension i: $\big||\operatorname{std}(\mathbf{e}_{u_j}) - \operatorname{std}(\mathbf{e}_n)| - |\operatorname{std}(\mathbf{e}_{u_j}) - \operatorname{std}({\mathbf{e}'}_n)|\big|$.

f_5 : The absolute difference between the cosine similarities before and after zeroing dimension i: $|\cos(\mathbf{e}_{u_j}, \mathbf{e}_n) - \cos(\mathbf{e}_{u_j}, {\mathbf{e}'}_n)|$.

Here, $\operatorname{mean}(\mathbf{e}) = \frac{1}{d}\sum_{t=1}^{d} \mathbf{e}^{(t)}$ is the mean of all dimensions of vector $\mathbf{e}$, and $\operatorname{std}(\mathbf{e})$ the standard deviation.

These features are concatenated ('$||$') as $\mathbf{z}_p$. We denote the feature preparation procedure above as $f_p(\cdot)$:

$$\mathbf{z}_p^i = f_p(e_{u_j}, e_n, i) = f_1||f_2||f_3||f_4||f_5, \tag{9}$$

Features f_1 and f_2 provide a basis on whether the i-th dimension is irrelevant; f_3 and f_4 capture the changes in statistical properties, reflecting the impact of dimension i; f_5 measures the change in cosine similarity after zeroing dimension i, preventing the direction of the user embedding from being dominated by a large-scale irrelevant noise feature.

Policy and action. The policy network maps $\mathbf{z}_p^i$ to a probability distribution and is implemented as:

$$p_i = \operatorname{sigmoid}(\omega \cdot \operatorname{ReLU}(\mathbf{W}\mathbf{z}_p^i + \mathbf{b})), \tag{10}$$

where $\mathbf{W}$ is a learnable weight matrix, $\mathbf{b}$ is a learnable bias, and ω is a learnable weight vector. A binary masking action $\mathbf{m} = [m_1, \ldots, m_d] \in \{0, 1\}^d$ is sampled from the policy:

$$m_i = \begin{cases} 1 & if\ p_i \geq \epsilon \\ 0 & otherwise \end{cases}, \tag{11}$$

where ϵ is sampled from $\mathcal{U}(0, 1)$. The Hadamard product of $\mathbf{m}$ and $\mathbf{e}_n^l$ filters out harmful dimensions.

Reward and loss. The reward is defined as the difference of the binary cross-entropy (BCELoss) of real user-item interactions ($r \in \mathbf{R}$) and predicted possibilities before (y) and after (y') zeroing dimension i. We denote the RL reward as ρ:

$$\rho = \operatorname{BCELoss}(r, y) - \operatorname{BCELoss}(r, y'). \tag{12}$$

To obtain the optimal parameters of the policy network, we minimize the following loss function:

$$\mathcal{L} = -\rho \cdot \log \pi(\mathbf{m} \mid \mathbf{z}_p^i); \pi(\mathbf{m} \mid \mathbf{z}_p^i) = \prod_{i=1}^{d} p_i^{m_i}(1 - p_i)^{1-m_i}. \tag{13}$$

We use the pre-trained policy agent to filter source-domain knowledge at target-domain training, detailed in Section IV-D.

b) Graph Expansion: After source-domain training and the preparation of policy agent, we feed $\mathbf{e}_{u_j}^l$ and $\mathbf{h}_l^{S_i,j}$ into the policy agent to obtain the masking action vector (Eq. 9-11) and hence the knowledge matrix ${\mathbf{H}'}^{S_i,j}$ after filtering out harmful dimensions. The rows of ${\mathbf{H}'}^{S_i,j}$ are calculated as:

$${\mathbf{h}'}_l^{S_i,j} = \mathbf{m} \odot \mathbf{h}_l^{S_i,j}, \quad m_i = \begin{cases} 1 & if\ p_i \geq 0.5 \\ 0 & otherwise \end{cases}, \tag{14}$$

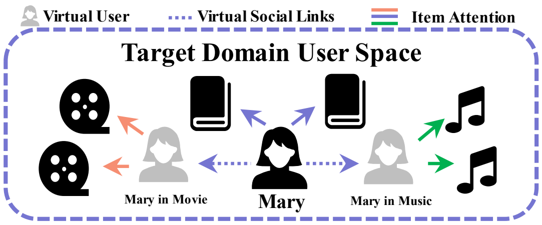

Fig. 3. Target domain graph expansion. The virtual users are constructed with source-domain knowledge embeddings of the `Movie` and `Music` domains.

where $\odot$ represents the Hadamard product. To simplify notation, we continue to use $\mathbf{h}_l^{S_i,j}$ and $\mathbf{H}^{S_i,j}$ to denote the filtered knowledge vector and matrix, respectively.

Next, the task is to incorporate the knowledge matrices $\mathbf{H}^{S_i,j}$ into the target-domain embedding of user u_j. FEDMCDR takes an attention-based approach. We expand the local graph of each user u_j by adding M *virtual users*. Each virtual user represents $\mathbf{H}^{S_i,j}$ from the same user u_j of source domain $\mathcal{D}^{S_i}$, as shown in Fig. 3. The idea is that these "same" user nodes have implicit social relationships with u_j, as reflected by the virtual social edges. The message propagation between them facilitates knowledge transfer. Since $\mathbf{H}^{S_i,j}$ comprises the $L-1$ intermediate user embeddings produced by $L-1$ GAT layers of a source domain, the l-th row, $\mathbf{h}^{S_i,j}$, can be directly incorporated into the computation of the l-th layer of the target-domain GAT. Thus, $\mathbf{H}^{S_i,j}$ functions the same as an actual user node, facilitating cross-domain knowledge transfer through message propagation. To strengthen the virtual social relationship, we incorporate a *social regularization* [26] term:

$$\mathcal{L}_s = \sum_{l=1}^{L} \|\mathbf{e}_{u_j}^l - \frac{\sum_{i=1}^{M} \text{sim}(\mathbf{e}_{u_j}^l, \mathbf{h}_l^{S_i,j}) \cdot \mathbf{h}_l^{S_i,j}}{\sum_{i=1}^{M} \text{sim}(\mathbf{e}_{u_j}^l, \mathbf{h}_l^{S_i,j})}\|^2, \tag{15}$$

where $\text{sim}(\cdot)$ calculates the cosine similarity.

The expanded graph with virtual users and social edges forms a heterogeneous structure with items of different types, as shown in Fig. 3. This heterogeneous graph is built under federated settings, facilitating cross-domain knowledge transfer while maintaining strong privacy guarantees.

Model heterogeneity. We considered all GATs from different domains to have the same number of layers for simplicity. In reality, the GATs used by different organizations may not have the same layer structure. Our method can be extended to such scenarios. When the number of layers of a source-domain GAT is greater than that of the target-domain GAT, we can discard the redundant layers. When the number of layers of a source-domain GAT is smaller, we still keep the virtual nodes for all layers of the target-domain GAT. The higher-layer embeddings of such nodes come from aggregations of the previous layers as part of the GAT training process.

D. Target Domain Training

The target-domain model $\mathcal{G}^{M+1}$ is trained on $|U|$ expanded local graphs on each client device, each of which has one user node, $|N_u|$ associated item nodes, and M virtual user nodes. It generates and assigns attention coefficients to both the actual and virtual users. The attention coefficients of M virtual users can be interpreted as M domain attention coefficients, which allow $\mathcal{G}^{M+1}$ to focus on domains that transfer positive knowledge. Embedding $\mathbf{h}_l^{S_i,j}$ is incorporated into the message propagation of GAT layer l, as a neighbor node of u_j.

$$\mathbf{e'}_{u_j}^l = \alpha_{u_j u_j}^l \mathbf{e}_{u_j}^l + \sum_{v \in N(u)} \alpha_{u_j v}^l \mathbf{e}_v^l + \sum_{i=1}^{M} \alpha_i^l \mathbf{h}_l^{S_i,j}, \tag{16}$$

where $\alpha_{u_j u_j}^l$, $\alpha_{u_j v}^l$, and α_i^l denote the attention coefficients of u_j towards itself, item v, and virtual user i, respectively.

We use inner product to estimate the interaction scores $\mathbf{R}_{u_j v}^{M+1}$ between user u_j and item v for the target domain:

$$y_{u_j v} = \text{sigmoid}(\mathbf{e'}_{u_j}^L \cdot \mathbf{e}_v), \tag{17}$$

We use the following loss function for $\mathcal{G}^{M+1}$:

$$\mathcal{L}^T = \text{BCELoss}(r_{u_j v}, y_{u_j v}) + \frac{\lambda}{2}\mathcal{L}_s, \tag{18}$$

where λ is a hyper-parameter, $\text{BCELoss}(\cdot)$ is the binary cross-entropy loss, and $\mathcal{L}_s$ is the social regulation term in Eq. 15.

Overall training pipeline. (i) Each of the $M+1$ domains independently and in parallel trains its own GAT model under horizontal FL. (ii) Based on the target domain GAT trained in Step (i), we pre-train the dimensional-filtering policy agent to avoid the apparent circular dependency. (iii) We sequentially run the knowledge preparation and positive knowledge activation modules for the training of the target domain model.

During this process, the training of the source and target domains is decoupled, thereby isolating the target domain from potential source-side fluctuations. Furthermore, this staged training strategy, coupled with weight freezing, establishes a stable representation space for the policy agent, effectively mitigating both training instability and the cold-start problem.

Discussion. Our design has two advantages: (i) the attention coefficients can vary across source domains, allowing the target model to focus on positive source-domain knowledge, and (ii) the attention coefficients are generated automatically, eliminating human efforts such as feature engineering.

In this extension, we remove the fine-tuning stage for $\mathcal{G}^{M+1}$ which was performed after graph expansion and target-domain training in the earlier version [10]. This is because the fine-tuning stage employs a training objective inconsistent with that of target-domain training, leading to negative impacts to the model accuracy as empirically shown in Section VII-E1.

Overall, our approach is complex yet highly scalable. Components such as source-domain knowledge and target-domain policy agents are trained once and reused. We report a detailed cost analysis in Section VI.

V. BOUNDING NEGATIVE TRANSFER

Most studies on CDR show their model capability to mitigate negative transfer through empirical accuracy of the target-domain model. Such accuracy results mix negative and positive knowledge from the source domains. The key challenge lies in separating and measuring negative knowledge. We fill this gap and make the first attempt to model negative transfer by quantifying the intensity of negative knowledge and its impact.

We first derive, in Section V-A, an upper bound on the impact of negative knowledge under *direct*, unprotected cross-domain transfer (i.e., a bound that applies to GAT-based federated CDR that transfers source-domain embeddings without defense mechanisms). We then show, in Sections V-B, that the two core modules of FEDMCDR provably tighten this bound.

Assumption 1 (Worst-case knowledge transfer): In order to simplify and facilitate the theoretical derivation, we assume a worst-case scenario where the source domains only transfer negative knowledge. We represent the negative knowledge of source domain i at layer l of an L-layer GAT ($0 < l \leq L$) as $\mathbf{k}_l^{S_i,j} \in \mathbb{R}^d$, where each dimension follows the Gaussian distribution $\mathcal{N}(\mu_i, \sigma_i^2)$.
We set $\mu_i = 0$, as any mean offset can be eliminated through alignment of mean values between source- and target-domain embeddings. The standard deviation σ_i quantifies the intensity of negative knowledge; we use $\sigma := \max_i \sigma_i$ to simplify notation.

A. Bounded Impact on Target-Domain under Direct Transfer

Unlike single-shot corruption (e.g., methods based on matrix factorization), negative knowledge in GAT is injected at *every* propagation layer. Based on Assumption 1 (i.e., $\mathbf{h}_l^{S_i,j} = \mathbf{k}_l^{S_i,j}$) and Eq. 16, we can derive the user embedding of layer l:

$$\mathbf{e'}_{u_j}^{l} = \alpha_{u_j u_j}^{l} \mathbf{e}_{u_j}^{l} + \sum_{v \in N(u)} \alpha_{u_j v}^{l} \mathbf{e}_{v}^{l} + \underbrace{\sum_{i=1}^{M} \alpha_i^l \, \mathbf{k}_l^{S_i,j}}_{noise-dependent}, \tag{19}$$

where $\mathbf{e'}_{u_j}^{l}$ is the user embedding output by the l-th layer; $\mathbf{e}_{u_j}^{l}, \mathbf{e}_{v}^{l}$ represent the embeddings of user u_j and item v at the l-th layer, respectively; and $\alpha_{u_j u_j}^{l}$, $\alpha_{u_j v}^{l}$, and α_i^l are the attention coefficients of user node u_j towards itself, item v, and virtual user i, respectively. Negative knowledge injected at an early layer $l < L$ does not vanish and is carried forward through subsequent layers and continues to contribute to the final-layer embedding $\mathbf{e'}_{u_j}^{L}$ used for prediction (cf. Eq. 17).

Consistent with the aforementioned worst-case simplification (i.e., only negative knowledge is transferred), we track the chain of self-loops, $u_j^l \to u_j^{l+1} \to \cdots \to u_j^L$, of L-layer GAT, which is the shortest, and empirically the highest-weight, path in GAT propagation.

Lemma 1 (Cumulative propagation factor): Define the layer-l-to-final-layer decay factor:

$$c_l \;:=\; \prod_{m=l+1}^{L} \alpha_{u_j u_j}^{m}, \qquad c_L := 1. \tag{20}$$

Then $c_l \in (0, 1]$ for all l's, and the noise-dependent component of $\mathbf{e'}_{u_j}^{L}$ along the propagation path is:

$$\sum_{l=1}^{L} c_l \sum_{i=1}^{M} \alpha_i^l \, \mathbf{k}_l^{S_i,j}. \tag{21}$$

Proof: By induction on the recursion $\mathbf{e}_{u_j}^{l+1} = \mathbf{e'}_{u_j}^{l}$: The layer-$l$ noise term $\sum_i \alpha_i^l \mathbf{k}_l^{S_i,j}$ enters $\mathbf{e}_{u_j}^{l+1}$ directly, which is then scaled by $\alpha_{u_j u_j}^{l+1}$ when propagating into $\mathbf{e}_{u_j}^{l+2}$ via the self-loop term, and so on up to layer L; the cumulative scaling is exactly $\prod_{m=l+1}^{L} \alpha_{u_j u_j}^{m} = c_l$. ■

We define the *cumulative propagation factor*:

$$C_L \;:=\; \sum_{l=1}^{L} c_l \;\in\; (0, L]. \tag{22}$$

Theorem 2 (Bounded impact under direct transfer): Substituting Eq. (21) into Eq. (17) and following the intensify argument (detailed next in Section V-C), the negative factor γ that determines the impact of negative transfer satisfies:

$$\log \gamma \;=\; \Big(\sum_{l=1}^{L} c_l \sum_{i=1}^{M} \alpha_i^l \, \mathbf{k}_l^{S_i,j} \Big) \cdot \mathbf{e}_v. \tag{23}$$

Moreover,

$$\mathbb{E}[\gamma_{\text{raw}}] \;=\; \exp\!\Big(\tfrac{1}{2} \|\mathbf{e}_v\|_1^2 \, \sigma^2 \, C_L^2 \Big). \tag{24}$$

Proof: Given negative knowledge $\mathbf{k}_l^{S_i,j}$ in the form of random Gaussian noise, we first substitute Eq. 21 into Eq. 19 and let $l = L$ to represent the final user embedding $\mathbf{e'}_{u_j}^{L}$ with cumulated negative knowledge.

$$\mathbf{e'}_{u_j}^{L} = \alpha_{u_j u_j}^{L} \mathbf{e}_{u_j}^{L} + \sum_{v \in N(u)} \alpha_{u_j v}^{L} \mathbf{e}_{v}^{L} + \sum_{l=1}^{L} c_l \sum_{i=1}^{M} \alpha_i^L \, \mathbf{k}_l^{S_i,j}. \tag{25}$$

Then, we substitute Eq. 19 into Eq. 16 to obtain the representation of $y_{u_j v}$ as a function of negative knowledge $\mathbf{k}_l^{S_i,j}$:

$$\begin{aligned} y_{u_j v} &= \frac{1}{1 + e^{-(\alpha_{u_j u_j}^{L} \mathbf{e}_{u_j}^{L} + \sum_{v \in N(u)} \alpha_{u_j v}^{L} \mathbf{e}_{v}^{L} + \sum_{l=1}^{L} c_l \sum_{i=1}^{M} \alpha_i^L \mathbf{k}_l^{S_i,j}) \cdot \mathbf{e}_v}} \\ &\quad \text{Separating the term with } \mathbf{k}_l^{S_i,j} \text{ and denoting it as } \gamma^{-1}: \\ &= \frac{1}{1 + e^{-(\alpha_{u_j u_j}^{L} \mathbf{e}_{u_j}^{L} + \sum_{v \in N(u)} \alpha_{u_j v}^{L} \mathbf{e}_{v}^{L}) \cdot \mathbf{e}_v} \cdot \underbrace{e^{-\sum_{l=1}^{L} c_l \sum_{i=1}^{M} \alpha_i^L \mathbf{k}_l^{S_i,j} \cdot \mathbf{e}_v}}_{\gamma^{-1}}} \\ &= \frac{1}{1 + \gamma^{-1} \cdot e^{-(\alpha_{u_j u_j}^{L} \mathbf{e}_{u_j}^{L} + \sum_{v \in N(u)} \alpha_{u_j v}^{L} \mathbf{e}_{v}^{L}) \cdot \mathbf{e}_v}}. \end{aligned} \tag{26}$$

Here, **only γ contains negative knowledge $\mathbf{k}_l^{S_i,j}$, reflecting its impact on the model.** We call it the *negative factor*.

We *intensify* the effect of negative knowledge to derive an upper bound $\tilde{y}_{u_j v}$ on the prediction:

$$\begin{aligned} y_{u_j v} \xrightarrow{\textbf{intensify}} \tilde{y}_{u_j v} &= \gamma \cdot \frac{1}{1 + e^{-(\alpha_{u_j u_j}^{L} \mathbf{e}_{u_j}^{L} + \sum_{v \in N(u)} \alpha_{u_j v}^{L} \mathbf{e}_{v}^{L}) \cdot \mathbf{e}_v}} \\ &= \gamma \cdot y_{u_j v}. \end{aligned} \tag{27}$$

Here, "intensify" means to amplify the impact of negative knowledge, i.e., to amplify the impact of the negative factor γ on $\tilde{y}_{u_j v}$. Compared to $y_{u_j v}$, $\tilde{y}_{u_j v}$ is more sensitive to variations of γ (i.e., negative knowledge $\mathbf{k}_l^{S_i,j}$), and thus exhibits a larger variation under negative transfer, providing an upper bound on its impact. We will show in Section V-C that the intensify operation amplifies the impact of negative knowledge.

Eq. 27 shows that the impact of negative knowledge on the predictions is upper bounded by γ. Now the core problem is transformed to studying the impact of γ. For the negative factor γ, we further analyze the exponential part of γ:

$$\log\gamma \;=\; \Big(\sum_{l=1}^{L} c_l \sum_{i=1}^{M} \alpha_i^l\, \mathbf{k}_l^{S_i,j}\Big)\cdot \mathbf{e}_v \tag{28}$$

$$< \sum_{l=1}^{L} c_l\, k_{\max}\sum_{i=1}^{M} \alpha_i^L \sum_{t\in d} \mathbf{e}_v^{(t)}$$

$$< \|\mathbf{e}_v\|_1\, k_{\max}\sum_{l=1}^{L} c_l \underbrace{\sum_{i=1}^{M}\alpha_i^L}_{<1}$$

$$< C_L\,\|\mathbf{e}_v\|_1\, k_{\max}, \tag{29}$$

where d is the dimension of embeddings, $\mathbf{e}_v^{(t)}$ denotes the t-th dimension of $\mathbf{e}_v$, and $k_{\max}$ is the maximum realized negative-knowledge magnitude. Since $k_{\max}$ follows the normal distribution, $\mathbf{e}_v$ is fixed, and $C_L \in (0, L]$ is governed by Lemma 1 regardless of noise realization, we have $\log\gamma \sim \mathcal{N}(0, \|\mathbf{e}_v\|_1^2\sigma^2 C_L^2)$. Eq. (24) follows directly from the mean of a log-normal random variable. ■

Eq. (24) shows that negative-transfer risk grows with C_L: deeper propagation and larger self-loop attentions (weaker filtering of one's own historical state) compound the effect of negative knowledge injected at every layer.

B. Bound Reduction via Adaptive Mapping and Positive Knowledge Activation

Theorem 2 bounds the impact of *directly transferred* negative knowledge, injected at every layer. We now show that the two core modules of FEDMCDR, the adaptive mapping function $f_A^i(\cdot)$ and the dimensional filter, tighten this bound.

Lemma 2 (Saturation via adaptive mapping): Let $z_{i,t} = \big(W_i\mathbf{k}_l^{S_i,j} + \mathbf{b}_i\big)_t$ and $h_l^{(t),S_i,j} = k_i\tanh(z_{i,t})$ be as defined by the adaptive mapping in Eq. 8, applied independently at every layer $l = 1,\ldots,L$. Then for every dimension t, every layer l, and every source domain i,

$$\big|h_l^{(t),S_i,j}\big| < k_i, \tag{30}$$

regardless of the magnitude of the input noise $k_{l,t}^{S_i,j}$ (i.e., σ_i), and regardless of the layer l at which it is injected.

Proof: Immediate from $\tanh(x) \in (-1, 1)\ \forall x \in \mathbb{R}$, applied independently at each layer. ■

This is the key structural difference from Theorem 2: raw negative knowledge $\mathbf{k}_l^{S_i,j}$ is unbounded and its per-layer contribution to C_L (Eq. (22)) grows linearly with σ_i, whereas after adaptive mapping, the *per-layer* magnitude is deterministically capped at k_i, independent of σ_i and l.

Assumption 2 (Effective filtering): Let $p_t = \Pr(m_t = 1)$ denote the retention probability of dimension t under the pre-trained policy (Eq. 11), and $\bar{p} = \max_t p_t$. Since the policy is trained to zero dimensions whose removal reduces the BCELoss (Eq. 12), for negative-knowledge-dominated inputs we have $\bar{p} < 1$ in expectation (empirically verified by the ablation FEDMCDR-M1 in Section VII-E, where removing the filter strictly hurts accuracy).

Theorem 3 (Bounded impact under FEDMCDR): Let $\mathbf{h}_l'^{S_i,j} = \mathbf{m}\odot\mathbf{h}_l^{S_i,j}$ be the transferred knowledge after both modules at layer l, and let

$$X \;=\; \sum_{l=1}^{L} c_l \sum_{i=1}^{M}\alpha_i^l\,\mathbf{h}_l'^{S_i,j}\cdot\mathbf{e}_v \tag{31}$$

be the exponent in γ (the multi-layer counterpart of the exponent in Theorem 2). Under Lemma 2 and Assumption 2,

$$\mathbb{E}[\gamma_{\text{FEDMCDR}}] \;\le\; \exp\!\Big(\tfrac{1}{2}\big(\|\mathbf{e}_v\|_1\,k_{\max}\,\bar{p}\,C_L\big)^2\Big);\, k_{\max} = \max_i k_i \tag{32}$$

Proof: By Lemma 2 and Assumption 2,

$$|X| \;\le\; \|\mathbf{e}_v\|_1\,k_{\max}\,\bar{p}\sum_{l=1}^{L} c_l \underbrace{\sum_{i=1}^{M}\alpha_i^l}_{<1} \;<\; \|\mathbf{e}_v\|_1\,k_{\max}\,\bar{p}\,C_L, \tag{33}$$

Thus, X is bounded in $[-B, B]$ with $B = \|\mathbf{e}_v\|_1 k_{\max}\bar{p}\,C_L$. Since $\mathbf{k}_l^{S_i,j}$ has zero mean, $\tanh(\cdot)$ is an odd function, and c_l, α_i^l are non-random scaling constants given the trained model, X is zero-mean. By Hoeffding's Lemma, a zero-mean random variable bounded in $[-B, B]$ satisfies $\mathbb{E}[e^X] \le e^{B^2/2}$. Substituting $B = \|\mathbf{e}_v\|_1 k_{\max}\bar{p}\,C_L$ gives Eq. (32). ■

Corollary 1 (Provable tightening relative to direct transfer, depth-invariant condition): Comparing Eq. (32) with Eq. (24) for the same target-domain backbone (same L, same realized attention structure, hence the same C_L – matching exactly how the ablations FEDMCDR-T1/M1 are implemented: identical architecture, only the two modules toggled),

$$\begin{aligned}\mathbb{E}[\gamma_{\text{FEDMCDR}}] < \mathbb{E}[\gamma_{\text{raw}}] \;&\iff\; k_{\max}\bar{p}\,C_L < \sigma\,C_L\\ &\iff\; k_{\max}\bar{p} < \sigma.\end{aligned} \tag{34}$$

Proof: Both exponents in Eqs. (24) and (32) are monotonically increasing functions of their respective scale factors, σC_L and $k_{\max}\bar{p}\,C_L$. Hence $\mathbb{E}[\gamma_{\text{FEDMCDR}}] < \mathbb{E}[\gamma_{\text{raw}}]$ if and only if $k_{\max}\bar{p}\,C_L < \sigma\,C_L$. Since $C_L > 0$, $k_{\max}\bar{p} < \sigma$. ■

Factor C_L *cancels exactly* from both sides of Eq. (34), which indicates that FEDMCDR's capability to tighten the bound is **invariant to network depth** L and to the specific attention values realized in c_l. It depends only on the saturation cap $k_{\max}$, the filter retention rate $\bar{p}$, and the noise intensity σ. Since k_i is initialized at the scale of the target-domain embedding range (e.g., $k_i = 1$ for $\mathcal{U}(-1, 1)$ initialization) and $\bar{p} < 1$, the left-hand side of Eq. (34) is $O(1)$, while $\sigma \in \{1, 2, 4\}$ in our negative-knowledge experiments (Section VII-C) is by construction at least this large; the condition therefore holds throughout our experimental range.

While the *condition* is depth-invariant, the *absolute gap* between the two bounds scales with C_L^2 (both exponents in Eqs. (24) and (32) scale multiplicatively with C_L^2). As propagation depth or self-loop weight increases, the benefit of FEDMCDR's mechanisms is *amplified*, not diminished. Moreover, as $\sigma \to \infty$, $\mathbb{E}[\gamma_{\text{raw}}] \to \infty$ while $\mathbb{E}[\gamma_{\text{FEDMCDR}}]$ remains bounded by a constant independent of σ. This formally explains the empirical observation in Table V that FEDMCDR's accuracy degrades far more gracefully than all baselines as the injected negative-knowledge intensity σ increases, and is consistent with the ablation results in Fig. 4

showing that removing either module (FEDMCDR-T1, which drops adaptive mapping; FEDMCDR-M1, which drops the RL filter) increases sensitivity to negative knowledge.

C. Proof of the Intensify Operation

For simplicity, we denote y as the original prediction $y_{u_j v}$ and y' as the prediction $\tilde{y}_{u_j v}$ after the intensify operation, for a generic exponent $\log \gamma$ of the form derived in Section V-A. We define the variation of negative knowledge as $\delta\gamma$ and the variation of the prediction as δy. The relative sensitivity is:

$$S = \frac{\gamma}{y} \cdot \frac{\delta y}{\delta \gamma} = \frac{\gamma}{y} \cdot \frac{\partial y}{\partial \gamma}. \tag{35}$$

Theorem 4 (Validity of the intensify operation): Let $t = -\big(\alpha^L_{u_j u_j} \mathbf{e}^L_{u_j} + \sum_{v \in N(u)} \alpha^L_{u_j v} \mathbf{e}^L_v\big) \cdot \mathbf{e}_v$. The relative sensitivity before the intensify operation, S_1, and after the intensify operation, S_2, satisfy $S_1 < S_2$ for any finite γ, t, such that $\tilde{y}_{u_j v} = \gamma \cdot y_{u_j v}$ is a valid upper bound on the impact of negative knowledge.

Proof: Before the intensify operation,

$$S_1 = \frac{\gamma}{y} \cdot \frac{\partial y}{\partial \gamma} = \gamma \cdot (1 + \gamma^{-1} e^{\mathbf{t}}) \cdot \frac{e^{\mathbf{t}}}{\gamma^2 (1 + \gamma^{-1} e^{\mathbf{t}})^2} = \frac{e^{\mathbf{t}}}{\gamma + e^{\mathbf{t}}} \tag{36}$$

After the intensify operation,

$$S_2 = \frac{\gamma}{y'} \cdot \frac{\partial y'}{\partial \gamma} = \gamma \cdot \gamma^{-1} (1 + e^{\mathbf{t}}) \cdot \gamma^{-2} \cdot \frac{e^{\mathbf{t}}}{\gamma^{-2} (1 + e^{\mathbf{t}})^2} = 1. \tag{37}$$

Since $\gamma, e^t > 0$, we have $S_1 = \frac{e^t}{\gamma + e^t} < 1 = S_2$ for any finite γ, t. Thus, the intensify operation strictly amplifies sensitivity to γ, confirming that $\tilde{y}_{u_j v}$ is a valid upper bound – for any $\log \gamma$ of the linear-in-noise form used throughout this section, single- or multi-layer alike. ■

VI. Cost Analysis

We analyze the cost of FEDMCDR in four dimensions: computation, communication, storage, and inference. The symbols used in the analysis are summarized in Table I.

TABLE I
NOTATION

Symbol	Meaning
M	number of source domains
L	number of GAT propagation layers
d	embedding dimensionality
$\lvert U \rvert$	number of users
$\bar{n}$	average number of item neighbors per user
R_S, R_T	number of federated training rounds

a) Computation Cost. The source-domain models $\mathcal{G}^1, \ldots, \mathcal{G}^M$ are each trained independently under standard horizontal FL. This costs $O(L\bar{n}d)$ per client per round, which is at the same order as training a single federated GAT recommender. Since the M source domains are mutually independent, this training is **parallel** across domain servers and, once completed, $\mathcal{G}^1, \ldots, \mathcal{G}^M$ are frozen and can be reused for all subsequent target-domain training rounds. The policy agent is also pretrained once, using the target domain's own frozen GAT (which is trained parallel to the training of the source domains and costs $O(L\bar{n}d)$). Its cost is that of one federated training run of a small MLP and is independent of M, since its input features f_1–f_5 are domain-agnostic statistics rather than source-domain-specific parameters. On the client side, at every round of the vertical stage, DP noise injection (Eq. 6) costs $O(Ld)$ and the adaptive mapping (Eq. 8) costs $O(Ld^2)$ per domain, giving $O(MLd^2)$ summed over M domains; dimensional filtering (Eq. 9–14) costs $O(ML\bar{n}d)$, dominated by feature preparation, since the policy MLP itself performs $O(1)$ work per dimension. Finally, target-domain training over the expanded graph (Eq. 16) costs $O(L(\bar{n} + M)d)$ per client per round. In practice $M \ll \bar{n}$ (up to 7 domains vs. tens to hundreds of item neighbors per user in our datasets).

Thus, the online, recurring computation overhead of FEDMCDR over a single-domain baseline is a small multiplicative factor, $O(1 + M/\bar{n})$. The more expensive $O(MLd^2)$ and $O(ML\bar{n}d)$ terms from knowledge preparation and positive activation are paid once per client (not per round) and can be cached, since the underlying source-domain knowledge does not change between target-domain training rounds. b) Communication Cost. Source-domain training communicates $O(Ld)$ per client per round. This happens M times, once per source domain, again in parallel across independent domain servers, so it does not multiply the wall-clock communication rounds required. Critically, the knowledge preparation and positive activation modules (Sections IV-B and IV-C) execute entirely on client devices using the source models already distributed during source-domain training. This adds zero additional communication. This is a structural property of the horizontal-vertical-horizontal design: unlike approaches that would need to query source-domain servers online for cross-domain embeddings at every round, no such round-trip is required here. Target-domain training then communicates $O(Ld)$ per client per round, again identical in order to a single-domain baseline, since the M virtual user nodes are locally materialized and not transmitted over the network. In short, the per-round communication cost of FEDMCDR is asymptotically the same as that of a single-domain federated recommender, independent of M.

c) Storage Cost. Each of the $M+1$ domain servers stores only its own item-embedding table and global GAT parameters, exactly as in $M+1$ independent single-domain federated recommenders; the multi-source design adds no server-side storage. On the client side, the only additional storage is (i) M small adaptive-mapping parameter sets $\{W_i, b_i, k_i\}$, costing $O(Md^2)$; given $d = 8$ and a mapping hidden size of 16 as used in our implementation, each mapping function has on the order of 2.8×10^2 parameters (roughly 1 KB), so even at $M = 7$ this totals under 10 KB per client; and (ii) M frozen source-domain attention vectors $\mathbf{a}^i \in \mathbb{R}^{2d}$, a few dozen bytes each. The policy agent is shared across all domains and its size is a small constant, independent of M. Client-side storage therefore grows only linearly in M with a very small constant factor, and does not depend on the number of federated training rounds or on $|U|$.

d) Inference Cost. At inference time, FEDMCDR performs a single forward pass through the expanded graph: $O(L(\bar{n} + M)d)$ per user, with no gradient computation. Two properties

TABLE II
STATISTICS ON THE AMAZON AND DOUBAN DATASETS.

Domains	$\|U\|$	$\|V\|$	$\|R\|$	Sparsity	Dataset
Home and Kitchen	5,539	114,416	215,503	0.0340%	Amazon
Books		115,740	200,103	0.0312%	
Tools and Home Improvement		52,638	92,618	0.0318%	
Kindle Store		102,093	192,494	0.0340%	
Clothing, Shoes and Jewelry		83,461	138,143	0.0299%	
Electronics		62,669	132,988	0.0383%	
Movies and TV		57,483	125,128	0.0393%	
Movie	1,364	20,953	820,234	0.0287%	Douban
Book		8,660	86,395	0.0007%	
Music		7,146	74,494	0.0008%	

make inference cheaper than training. First, the adaptive-mapping step is skipped at inference time. We transfer the DP-perturbed embeddings $\tilde{\mathbf{X}}^{S_i,j}$ directly rather than mapping them (Section IV-B), avoiding the $O(MLd^2)$ mapping cost. Second, since the M virtual-user embeddings and the trained policy agent are already cached on the client from the (offline) knowledge preparation and positive activation stages, inference requires no network communication. The per-request latency overhead relative to a single-domain federated GAT recommender is therefore limited to attending to M extra (locally cached) virtual neighbor nodes, i.e., the same small $O(1 + M/\bar{n})$ factor identified in the computation analysis above.

VII. EXPERIMENTS

We empirically answer the following questions: **(Q1)** How does FEDMCDR perform in recommendation quality in the target domain, comparing with SOTA models? **(Q2)** How does the intensity of negative transfer impact the recommendation quality of FEDMCDR? **(Q3)** How do the components of FEDMCDR contribute to its overall effectiveness? **(Q4)** Does FEDMCDR sustain robustness in more complex noise scenarios? **(Q5)** Does FEDMCDR sustain robustness under non-zero domain bias? **(Q6)** How does FEDMCDR perform in scenarios with partial user set overlap?

A. Experimental Setup

a) Datasets: We run experiments with two real-world datasets, the **Amazon** dataset (Amazon Reviews 2023 [27]) and the **Douban** dataset [28], as summarized in Table II.

We note that our conference paper [10] used Amazon Reviews 2018 [29] which is an earlier version of the Amazon dataset. Comparing with the 2018 version, the 2023 dataset has more domains (33 vs. 29) and more reviews in each domain. Overall, the 2023 dataset has 571.54 million reviews, which is 245.2% larger than the 2018 version.

As the number of overlapping users decreases with more domains, we focus on the top seven domains with the most user ratings to ensure a non-trivial number of overlapping users, i.e., 5,539. The overlapping users form the user set. Items from each domain that were rated by at least one user in the user set form the item set for that domain.

To study the impact of the number of domains, we create three dataset instances containing the first three, five, and seven domains as listed in Table II, denoted as **Amazon-3**, **Amazon-5**, and **Amazon-7**, respectively. We list the Tools and Home Improvement (Tools) as the third domain to make a "smaller" target domain. For the Douban dataset, we use the data from all three domains and filter out items with fewer than five user ratings following the literature [2], [10]. We iteratively treat the domains in **Amazon-3** (i.e., Home, Books, and Tools), and **Douban** (i.e., Movie, Book, and Music) as the target domain while the rest serving as the source domains.

b) Competitors: We compare with five models, including three SOTA federated CDR models using decentralized user data storage (cf. Section II): (i) **FedCT** [2], (ii) **FedCDR** [3], and (iii) **Fed-CLR** [7], as well as two adapted classic CDR models: (iv) **EMCDR** [18] is an embedding-mapping CDR framework, and (v) **PriCDR** [30] is a privacy-preserving CDR framework, which applies DP on rating matrices. We adapt **EMCDR** and **PriCDR** to horizontal FL following FedCT [2].

Besides, we compare with **FedGCDR** which is our earlier model from the conference paper [10].

c) Implementation Details: We set the number of propagation layers of GAT-base federated models across all domains to 2. The adaptive mapping function has one hidden layers of size 16. We set $\lambda = 0.01$ in the objective function $\mathcal{L}^T(\cdot)$. We set the embedding dimensionality to 8 for all domains and use We set the learning rate to 0.01 for GAT model training and 0.001 when pre-training the policy agent. We use AdamW as the optimizer with a weight decay coefficient of 1.

d) Evaluation Protocol: All experiments were run on machines with four NVIDIA RTX 3090 GPUs. We use the leave-one-out method which is widely used in recommender system evaluation [11]. The item that a user rated the last (in time) and 99 other randomly sampled unrated items form the test set. The remaining user-item interaction data is used for model training. We report the Hit Rate (**HR@k**) and the Normalized Discounted Cumulative Gain (**NDCG@k**) [31] of the top-k ranked items with $k = 5, 10$. Model performance is assessed by its ability to rank the true positive item higher than the 99 negatives, as measured by HR@k and NDCG@k.

We repeat each experiment for three times with different random seeds and report the average performance results.

B. Recommendation Performance (Q1)

1) Amazon Dataset: Table III reports model accuracy on the Amazon dataset, where we vary the number of domains $|M{+}1| \in \{3, 5, 7\}$ and iteratively treat Home, Books, and Tools as the target domain. Overall, FEDMCDR achieves the best results across all settings and all metrics, demonstrating its effectiveness in federated multi-source CDR.

When $|M{+}1| = 3$, FEDMCDR significantly outperforms all competitors on the three target domains. Compared with the strongest baseline FedGCDR, FEDMCDR improves HR@10 by 42.0% (0.3959 vs. 0.2788) on Home, 28.2% (0.3936 vs. 0.3070) on Books, and 37.6% (0.3509 vs. 0.2551) on Tools. Similar gains are observed in NDCG@5 and NDCG@10, indicating that FEDMCDR not only improves the hit probability but also ranks the relevant item higher.

As the number of domains increases (from 3 to 7), the changes in performance can vary. For Home (Amazon-7@Home) and Tools (Amazon-7@Tools), additional source

TABLE III
RECOMMENDATION ACCURACY ON AMAZON. THE BEST RESULT FOR EACH SETTING IS MARKED IN BOLD AND THE SECOND BEST IS UNDERLINED.

$\|M+1\|$	Model	Home				Books				Tools			
		H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10
3	EMCDR	0.1824 ±.0004	0.1307 ±.0001	0.2357 ±.0003	0.1479 ±.0001	0.2029 ±.0008	0.1522 ±.0005	0.2577 ±.0005	0.1698 ±.0004	0.1539 ±.0011	0.109 ±.0005	0.2134 ±.0008	0.1281 ±.0005
	PriCDR	0.1894 ±.0027	0.1350 ±.0014	0.2577 ±.0013	0.1570 ±.0004	0.2036 ±.0006	0.1526 ±.0009	0.2610 ±.0015	0.1712 ±.0006	0.1578 ±.0007	0.1109 ±.0004	0.2220 ±.0019	0.1316 ±.0008
	FedCT	0.1856 ±.0009	0.1326 ±.0003	0.2463 ±.0015	0.1522 ±.0005	0.1999 ±.0001	0.1498 ±.0002	0.2550 ±.0004	0.1676 ±.0001	0.1554 ±.0005	0.1099 ±.0002	0.2208 ±.0005	0.1310 ±.0002
	FedCDR	0.1595 ±.0062	0.1149 ±.0038	0.2045 ±.0081	0.1295 ±.0045	0.1613 ±.0118	0.1206 ±.0095	0.2056 ±.0101	0.1349 ±.0087	0.1255 ±.0047	0.0879 ±.0034	0.1715 ±.0022	0.1027 ±.0026
	Fed-CLR	0.0836 ±.0015	0.0548 ±.0011	0.1378 ±.0014	0.0721 ±.0006	0.0663 ±.0041	0.0418 ±.0023	0.1190 ±.0040	0.0586 ±.0020	0.0624 ±.0026	0.0388 ±.0024	0.1154 ±.0035	0.0556 ±.0028
	FedGCDR	0.2073 ±.0011	0.1449 ±.0007	0.2788 ±.0017	0.1681 ±.0006	0.2217 ±.0037	0.1606 ±.0024	0.3070 ±.0037	0.1880 ±.0023	0.1779 ±.0034	0.1207 ±.0020	0.2551 ±.0033	0.1457 ±.0017
	FEDMCDR	**0.3482** ±.0125	**0.2491** ±.0074	**0.3959** ±.0102	**0.2648** ±.0065	**0.3439** ±.0024	**0.2558** ±.0016	**0.3936** ±.0026	**0.2719** ±.0010	**0.2542** ±.0096	**0.1655** ±.006	**0.3509** ±.0079	**0.1970** ±.0055
5	EMCDR	0.1833 ±.0017	0.1312 ±.0010	0.2359 ±.0010	0.1481 ±.0008	0.2026 ±.0007	0.1521 ±.0003	0.2579 ±.0005	0.1699 ±.0001	0.1543 ±.0008	0.1091 ±.0005	0.2139 ±.0011	0.1283 ±.0004
	PriCDR	0.1895 ±.0009	0.1354 ±.0007	0.2610 ±.0006	0.1585 ±.0006	0.2050 ±.0018	0.1531 ±.0012	0.2610 ±.0014	0.1712 ±.0005	0.1583 ±.0013	0.1111 ±.0007	0.2212 ±.0005	0.1314 ±.0004
	FedCT	0.1858 ±.0008	0.1328 ±.0003	0.2462 ±.0008	0.1523 ±.0002	0.1996 ±.0003	0.1496 ±.0002	0.2553 ±.0010	0.1675 ±.0003	0.1558 ±.0003	0.1100 ±.0003	0.2205 ±.0014	0.1307 ±.0003
	FedCDR	0.1570 ±.0045	0.1138 ±.0024	0.2047 ±.0049	0.1293 ±.0024	0.1645 ±.0115	0.1225 ±.0088	0.2079 ±.0118	0.1365 ±.0089	0.1254 ±.0048	0.0874 ±.0043	0.1692 ±.0053	0.1015 ±.0045
	Fed-CLR	0.0641 ±.0055	0.0399 ±.0044	0.1159 ±.0046	0.0563 ±.0042	0.0639 ±.0024	0.0399 ±.0015	0.1143 ±.0040	0.0560 ±.0012	0.0603 ±.0011	0.0371 ±.0011	0.1169 ±.0041	0.0553 ±.0021
	FedGCDR	0.2043 ±.0013	0.1431 ±.0007	0.2747 ±.0020	0.1659 ±.0008	0.2244 ±.0042	0.1623 ±.0022	0.3070 ±.0053	0.1889 ±.0025	0.1705 ±.0007	0.1168 ±.0002	0.2440 ±.0014	0.1405 ±.0006
	FEDMCDR	**0.3384** ±.0074	**0.2433** ±.0065	**0.3845** ±.0057	**0.2583** ±.0058	**0.3535** ±.0058	**0.2652** ±.0054	**0.4034** ±.0044	**0.2813** ±.0049	**0.2464** ±.0048	**0.1622** ±.0044	**0.3307** ±.0037	**0.1897** ±.0038
7	EMCDR	0.1841 ±.0007	0.1321 ±.0001	0.2370 ±.0008	0.1492 ±.0008	0.2026 ±.0007	0.1521 ±.0003	0.2581 ±.0006	0.1699 ±.0003	0.1541 ±.0010	0.1089 ±.0005	0.2136 ±.0009	0.1280 ±.0002
	PriCDR	0.1920 ±.0012	0.1360 ±.0004	0.2622 ±.0010	0.1583 ±.0014	0.2039 ±.0012	0.1529 ±.0010	0.2612 ±.0011	0.1714 ±.0006	0.1583 ±.0012	0.1110 ±.0005	0.2211 ±.0007	0.1312 ±.0005
	FedCT	0.1856 ±.0002	0.1325 ±.0001	0.2464 ±.0001	0.1522 ±.0008	0.1993 ±.0009	0.1494 ±.0003	0.2545 ±.0019	0.1671 ±.0010	0.1554 ±.0006	0.1098 ±.0003	0.2208 ±.0008	0.1308 ±.0002
	FedCDR	0.1571 ±.0002	0.1141 ±.0001	0.2042 ±.0002	0.1296 ±.0028	0.1623 ±.0109	0.1211 ±.0092	0.2053 ±.0118	0.1349 ±.0095	0.1250 ±.0044	0.0879 ±.0030	0.1703 ±.0047	0.1025 ±.0030
	Fed-CLR	0.0793 ±.0101	0.0526 ±.0049	0.1397 ±.0055	0.0718 ±.0087	0.0655 ±.0034	0.0411 ±.0017	0.1213 ±.0043	0.0589 ±.0020	0.0650 ±.0076	0.0402 ±.0057	0.1170 ±.0072	0.0568 ±.0056
	FedGCDR	0.2009 ±.0103	0.1414 ±.0148	0.2721 ±.0132	0.1645 ±.0101	0.2235 ±.0052	0.1620 ±.0024	0.3071 ±.0047	0.1890 ±.0021	0.1684 ±.0015	0.1156 ±.0013	0.2408 ±.0008	0.1390 ±.0012
	FEDMCDR	**0.3239** ±.0060	**0.2327** ±.0046	**0.3708** ±.0067	**0.2479** ±.0045	**0.3462** ±.0046	**0.2578** ±.0027	**0.3961** ±.0025	**0.2739** ±.0021	**0.2456** ±.0060	**0.1614** ±.0043	**0.3260** ±.0017	**0.1876** ±.0029

TABLE IV
RECOMMENDATION ACCURACY ON DOUBAN.

Model	Movie				Book				Music			
	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10
EMCDR	0.4527 ±.0068	0.2635 ±.0037	0.6325 ±.0100	0.3239 ±.0049	0.2643 ±.0019	0.2019 ±.0011	0.3759 ±.0074	0.2373 ±.0022	0.2156 ±.0025	0.1468 ±.0019	0.3233 ±.0046	0.1818 ±.0025
PriCDR	0.4712 ±.0297	0.2675 ±.0048	0.6549 ±.0113	0.3286 ±.0117	0.2828 ±.0187	0.2175 ±.0097	0.3837 ±.0182	0.2515 ±.0044	0.2238 ±.0145	0.1527 ±.0079	0.3409 ±.0159	0.1896 ±.0084
FedCT	0.4556 ±.0037	0.2652 ±.0021	0.6378 ±.0045	0.3264 ±.0023	0.2639 ±.0021	0.2039 ±.0012	0.3811 ±.0023	0.2440 ±.0009	0.2181 ±.0023	0.1487 ±.0016	0.3277 ±.0021	0.1843 ±.0015
FedCDR	0.4610 ±.0711	0.2830 ±.0694	0.6268 ±.0220	0.3358 ±.0448	0.2582 ±.0153	0.1991 ±.0289	0.3196 ±.0164	0.2189 ±.0263	0.2170 ±.0266	0.1471 ±.0278	0.3127 ±.0373	0.1780 ±.0322
Fed-CLR	0.4503 ±.0256	0.2805 ±.0210	0.5925 ±.0221	0.3166 ±.0189	0.2053 ±.0175	0.1556 ±.0149	0.2740 ±.0202	0.1774 ±.0145	0.2025 ±.0256	0.1455 ±.0210	0.2806 ±.0221	0.1707 ±.0189
FedGCDR	0.4927 ±.0332	0.2949 ±.0090	0.6496 ±.0046	0.3470 ±.0157	0.2885 ±.0176	0.2213 ±.0082	0.3849 ±.0159	0.2522 ±.0068	0.2239 ±.0055	0.1549 ±.0044	0.3526 ±.0116	0.1945 ±.0067
FEDMCDR	**0.4966** ±.0111	**0.2994** ±.0011	**0.6514** ±.0090	**0.3488** ±.0056	**0.2906** ±.0060	**0.2271** ±.0057	**0.3864** ±.0113	**0.2561** ±.0084	**0.2251** ±.0115	**0.1556** ±.0109	**0.3602** ±.0023	**0.1966** ±.0079

domains contribute negatively and hurts the model accuracy. However, for Books (Amazon-7@Books), those domains help improve our model accuracy. These observations reveal the complexity of the impact between domains.

Overall, adding more source domains does not guarantee accuracy improvement in CDR, because it may increase domain heterogeneity and intensify negative transfer. FEDMCDR achieves the best results across all settings, while its accuracy could also decline with more source domains. This emphasizes the challenges brought by multiple source domains.

TABLE V
MODEL ACCURACY ON AMAZON@HOME WHEN VARYING NEGATIVE KNOWLEDGE INTENSITY.

$\|M+1\|$	Model	$\sigma=1$ H@5	$\sigma=1$ N@5	$\sigma=1$ H@10	$\sigma=1$ N@10	$\sigma=2$ H@5	$\sigma=2$ N@5	$\sigma=2$ H@10	$\sigma=2$ N@10	$\sigma=4$ H@5	$\sigma=4$ N@5	$\sigma=4$ H@10	$\sigma=4$ N@10
3	EMCDR	0.1815 ±.0007	0.1308 ±.0005	0.2354 ±.0007	0.1482 ±.0006	0.1826 ±.0009	0.1312 ±.0007	0.2351 ±.0006	0.1482 ±.0006	0.1818 ±.0003	0.1308 ±.0005	0.2350 ±.0010	0.1479 ±.0007
	PriCDR	0.0969 ±.0026	0.0682 ±.0021	0.1525 ±.0037	0.0859 ±.0022	0.0560 ±.0021	0.0333 ±.0011	0.1123 ±.0022	0.0514 ±.0011	0.0590 ±.0016	0.0360 ±.0017	0.1155 ±.0020	0.0543 ±.0020
	FedCT	0.1865 ±.0012	0.1330 ±.0005	0.2432 ±.0011	0.1513 ±.0003	0.1831 ±.0014	0.1313 ±.0007	0.2373 ±.0026	0.1488 ±.0007	0.1719 ±.0017	0.1235 ±.0020	0.2207 ±.0036	0.1392 ±.0025
	FedCDR	0.1150 ±.0007	0.0718 ±.0011	0.1735 ±.0034	0.0787 ±.0017	0.1038 ±.0007	0.0628 ±.0013	0.1729 ±.0014	0.0751 ±.0014	0.0921 ±.0023	0.0548 ±.0013	0.1575 ±.0017	0.0671 ±.0007
	Fed-CLR	0.0795 ±.0038	0.0514 ±.0032	0.1348 ±.0047	0.0690 ±.0033	0.0692 ±.0013	0.0437 ±.0005	0.1243 ±.0046	0.0613 ±.0009	0.0722 ±.0026	0.0466 ±.0017	0.1263 ±.0023	0.0639 ±.0016
	FedGCDR	0.2050 ±.0013	0.1434 ±.0006	0.2756 ±.0014	0.1665 ±.0002	0.2020 ±.0026	0.1417 ±.0014	0.2720 ±.0023	0.1646 ±.0004	0.2004 ±.0014	0.1409 ±.0013	0.2691 ±.0007	0.1633 ±.0010
	FEDMCDR	**0.3317** ±.0111	**0.2348** ±.0072	**0.3833** ±.0079	**0.2516** ±.0062	**0.2975** ±.0110	**0.2074** ±.0067	**0.3554** ±.0077	**0.2262** ±.0057	**0.2530** ±.0056	**0.1748** ±.0024	**0.3191** ±.0035	**0.1962** ±.0019
5	EMCDR	0.1823 ±.0014	0.1308 ±.0009	0.2346 ±.0010	0.1477 ±.0008	0.1816 ±.0015	0.1303 ±.0012	0.2333 ±.0013	0.1471 ±.0011	0.1811 ±.0019	0.1300 ±.0015	0.2334 ±.0023	0.1469 ±.0015
	PriCDR	0.1599 ±.0014	0.1155 ±.0009	0.2120 ±.0012	0.1322 ±.0010	0.0905 ±.0025	0.0621 ±.0017	0.1437 ±.0023	0.0791 ±.0011	0.0530 ±.0025	0.0316 ±.0014	0.1075 ±.0008	0.0490 ±.0010
	FedCT	0.1854 ±.0019	0.1320 ±.0009	0.2434 ±.0012	0.1508 ±.0007	0.1820 ±.0017	0.1303 ±.0014	0.2358 ±.0011	0.1477 ±.0009	0.1673 ±.0018	0.1209 ±.0012	0.2170 ±.0033	0.1370 ±.0013
	FedCDR	0.1143 ±.0020	0.0722 ±.0007	0.1737 ±.0021	0.0793 ±.0006	0.1032 ±.0027	0.0632 ±.0016	0.1721 ±.0017	0.0753 ±.0010	0.0887 ±.0023	0.0531 ±.0012	0.1590 ±.0016	0.0675 ±.0006
	Fed-CLR	0.0672 ±.0012	0.0430 ±.0004	0.1229 ±.0029	0.0608 ±.0013	0.0652 ±.0115	0.0402 ±.0077	0.1195 ±.0089	0.0575 ±.0068	0.0634 ±.0078	0.0397 ±.0060	0.1153 ±.0076	0.0563 ±.0059
	FedGCDR	0.2010 ±.0015	0.1406 ±.0004	0.2695 ±.0024	0.1628 ±.0008	0.1984 ±.0020	0.1398 ±.0009	0.2659 ±.0015	0.1617 ±.0006	0.1962 ±.0035	0.1390 ±.0017	0.2633 ±.0010	0.1609 ±.0008
	FEDMCDR	**0.3119** ±.0054	**0.2204** ±.0050	**0.3652** ±.0033	**0.2378** ±.0044	**0.2704** ±.0063	**0.1900** ±.0051	**0.3319** ±.0058	**0.2100** ±.0048	**0.2298** ±.0034	**0.1615** ±.0025	**0.2994** ±.0057	**0.1840** ±.0032
7	EMCDR	0.1837 ±.0008	0.1318 ±.0006	0.2365 ±.0002	0.1487 ±.0005	0.1852 ±.0001	0.1318 ±.0001	0.2363 ±.0002	0.1488 ±.0009	0.1839 ±.0006	0.1318 ±.0002	0.2360 ±.0002	0.1482 ±.0006
	PriCDR	0.1903 ±.0013	0.1349 ±.0012	0.2588 ±.0018	0.1566 ±.0012	0.1823 ±.0022	0.1289 ±.0012	0.2498 ±.0025	0.1493 ±.0005	0.1484 ±.0039	0.1030 ±.0017	0.2151 ±.0018	0.1240 ±.0013
	FedCT	0.1852 ±.0004	0.1327 ±.0007	0.2437 ±.0005	0.1517 ±.0004	0.1840 ±.0009	0.1316 ±.0004	0.2407 ±.0007	0.1500 ±.0001	0.1762 ±.0007	0.1261 ±.0002	0.2282 ±.0006	0.1428 ±.0001
	FedCDR	0.1257 ±.0001	0.0878 ±.0001	0.1798 ±.0001	0.0992 ±.0004	0.1161 ±.0024	0.0781 ±.0027	0.1758 ±.0004	0.0843 ±.0004	0.0951 ±.0024	0.0581 ±.0006	0.1598 ±.0026	0.0725 ±.0001
	Fed-CLR	0.0788 ±.0046	0.0516 ±.0025	0.1381 ±.0066	0.0702 ±.0027	0.0766 ±.0028	0.0484 ±.002	0.1357 ±.0021	0.0691 ±.0069	0.0726 ±.0075	0.0477 ±.0016	0.1329 ±.0123	0.0675 ±.0167
	FedGCDR	0.1962 ±.0100	0.1382 ±.0190	0.2670 ±.0158	0.1612 ±.0118	0.1944 ±.0113	0.1380 ±.0129	0.2632 ±.0112	0.1604 ±.0105	0.1935 ±.0149	0.1373 ±.0131	0.2600 ±.0173	0.1589 ±.0107
	FEDMCDR	**0.2883** ±.0048	**0.2054** ±.0044	**0.3428** ±.0061	**0.2231** ±.0047	**0.2397** ±.0057	**0.1694** ±.0047	**0.3092** ±.0059	**0.1918** ±.0050	**0.2078** ±.0040	**0.1454** ±.0030	**0.2759** ±.0035	**0.1675** ±.0028

2) Douban Dataset: Table IV reports model accuracy on the Douban dataset, which contains three smaller and denser domains (Movie, Book, and Music). FEDMCDR consistently outperforms all baselines, with more noticeable gains in the Music domain comparing with the strongest baseline FedGCDR (e.g., HR@10: 0.3602 vs. 0.3526). These results show that FEDMCDR can effectively exploit cross-domain signals and is robust for a denser recommendation scenario.

C. Ability to Mitigate Negative Transfer (Q2)

To evaluate model robustness against negative transfer under a controllable and comparable setting, we add negative knowledge sampled Gaussian distribution, i.e., a transferred knowledge embedding $\mathbf{x}$ is modified to be $\mathbf{x}' = \mathbf{x} + (r_1, r_2, \ldots, r_k)$, where r_i is randomly drawn from $\mathcal{N}(0, \sigma^2)$, $\sigma \in \{0, 1, 2, 4\}$. Here, $\sigma = 0$ corresponds to the baseline scenario without additional negative knowledge, while $\sigma = 4$ represents scenarios with strong negative knowledge.

Table V reports the results on Amazon@Home, where $|M+1| = 3, 5, 7$. As σ increases, all methods suffer, confirming that negative knowledge can harm CDR. FEDMCDR remains the most accurate, showing stronger robustness against noisy transferred knowledge. When $\sigma = 4$, the expected absolute value of the injected noise as $|E| = \sigma\sqrt{\frac{2}{\pi}} \approx 3.2$, which means each dimension of the transferred knowledge is expected to be affected by a noise term with a value of 3.2, creating an almost fully negative-transfer scenario. In this scenario, we can better understand the model's ability to mitigate negative transfer, and FEDMCDR shows strong resistance in this case. We omit results on the other domains for conciseness as the patterns are similar.

Across the experiments above, FEDMCDR outperforms the strongest baseline FedGCDR by up to 42.0% in HR@10 and 57.5% in NDCG@10 (Amazon-3@Home in Table III).

D. Macro-average Relative Improvement

To summarize the overall improvements of FEDMCDR, we report the *macro-average relative improvement* over the strongest baseline FedGCDR on HR@k and NDCG@k.

TABLE VI
RESULTS OF ADDITIONAL MODEL VARIANTS ON AMAZON-7@HOME.

Model	$\sigma = 0$				$\sigma = 1$				$\sigma = 2$				$\sigma = 4$			
	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10
FEDMCDR	**0.3239**	**0.2327**	**0.3708**	**0.2479**	**0.2883**	**0.2054**	**0.3428**	**0.2231**	**0.2397**	**0.1694**	**0.3092**	**0.1918**	**0.2078**	**0.1454**	**0.2759**	**0.1675**
	±.0060	±.0046	±.0067	±.0045	±.0048	±.0044	±.0061	±.0047	±.0057	±.0047	±.0059	±.0050	±.0040	±.0030	±.0035	±.0028
FEDMCDR-V1	0.3067	0.2211	0.3659	0.2403	0.2671	0.1916	0.3328	0.2128	0.2329	0.1653	0.3000	0.1869	0.2021	0.1430	0.2704	0.1648
	±.0181	±.0125	±.0167	±.0120	±.0132	±.0095	±.0138	±.0096	±.0030	±.0035	±.0035	±.0036	±.0034	±.0010	±.0016	±.0013
FEDMCDR-V2	0.1629	0.1116	0.2251	0.1318	0.1589	0.1092	0.2220	0.1296	0.1553	0.1054	0.2231	0.1274	0.1414	0.0963	0.2027	0.1161
	±.0031	±.0017	±.0027	±.0017	±.0028	±.0019	±.0036	±.0021	±.0022	±.0004	±.0036	±.0011	±.0014	±.0016	±.0036	±.0012
FEDMCDR-V3	0.2471	0.1647	0.3295	0.1915	0.2257	0.1521	0.3071	0.1785	0.2071	0.1423	0.2847	0.1674	0.1968	0.1379	0.2684	0.1611
	±.0065	±.0044	±.0037	±.0032	±.0072	±.0045	±.0032	±.0030	±.0028	±.0022	±.0024	±.0014	±.0023	±.0015	±.0007	±.0013

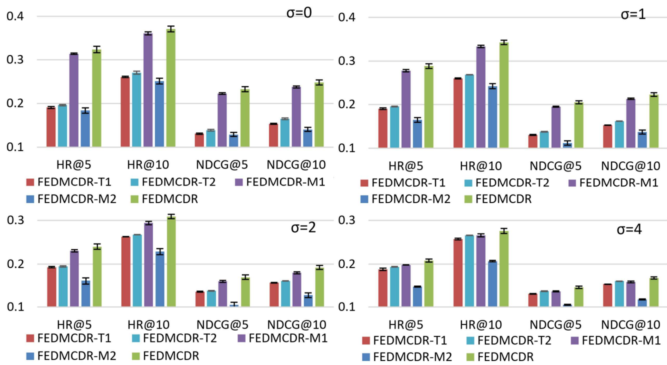

Fig. 4. Results of module ablation on Amazon-7@Home.

For Tables III to V, each table contains $\mathcal{T}$ experimental settings, where $\mathcal{T}$ is the product of the number of target domains, the number of different source-domain settings, and the number of different noise intensity settings. For each setting $t \in \mathcal{T}$ and each metric x (e.g., HR@10), we compute the relative improvement:

$$\Delta_{t_x} = \frac{\text{Score}_{\text{FEDMCDR}}(t, x) - \text{Score}_{\text{FedGCDR}}(t, x)}{\text{Score}_{\text{FedGCDR}}(t, x)}. \tag{38}$$

For example, in Table III ($\mathcal{T} = 3 \times 3 \times 1 = 9$), when $|M{+}1| = 3$ and the target domain is Home, the HR@10 scores are 0.3959 (FEDMCDR) and 0.2788 (FedGCDR), yielding $\Delta_{t_{\text{HR@10}}} = (0.3959 - 0.2788)/0.2788 = 42.0\%$. We then compute the macro-average of metric x across all settings by

$$\text{MA}_x = \frac{1}{\mathcal{T}_{\text{II}} + \mathcal{T}_{\text{III}} + \mathcal{T}_{\text{IV}}} \Big(\sum_{t \in \mathcal{T}_{\text{II}}} \Delta_{t_x} + \sum_{t \in \mathcal{T}_{\text{III}}} \Delta_{t_x} + \sum_{t \in \mathcal{T}_{\text{IV}}} \Delta_{t_x} \Big), \tag{39}$$

where $\mathcal{T}_{\text{II}}$, $\mathcal{T}_{\text{III}}$, and $\mathcal{T}_{\text{IV}}$ denote the numbers of experimental settings in Tables III, IV, and V, respectively.

Under this definition, FEDMCDR achieves macro-average relative improvements of 39.0%/39.5% and 25.4%/32.0% on HR@5/NDCG@5 and HR@10/NDCG@10, respectively.

E. Ablation Study (Q3)

1) Module Ablation: To study the contribution of each module in FEDMCDR, we implement four model variants, **FEDMCDR-T1**, **FEDMCDR-T2**, **FEDMCDR-M1**, and **FEDMCDR-M2**. FEDMCDR-T1 reverts to the MLP mapping design (Eq. 7) from our conference version (FedGCDR), and the mapping is applied during both training and prediction. It validates the knowledge preparation module. FEDMCDR-T2 uses adaptive mapping during both training and prediction. FEDMCDR-M1 removes the reinforcement learning (RL)-based dimension filtering, i.e., the source-domain embeddings are directly used. FEDMCDR-M2 adds model fine-tuning as in FedGCDR. These together validate the positive knowledge activation module.

We report results on Amazon-7@Home and omit the others due to space limit.

As shown in Fig. 4, FEDMCDR consistently outperforms all four model variants, validating the necessity of adaptive knowledge mapping, dimension-level knowledge filtering, and end-to-end training without additional fine-tuning, which are new designs of FEDMCDR. The fact that FEDMCDR-T1 and FEDMCDR-T2 are both outperformed by FEDMCDR confirms the necessity of adaptive mapping (i.e., using Eq. 8 instead of MLP as in FEDMCDR-T1) and its training-only usage (i.e., not using it for both training and inference as done in FEDMCDR-T2), respectively. FEDMCDR-M2 has the worst accuracy. It has an additional fine-tuning stage that freezes the GAT and trains only $\mathbf{e}'^{L}_{u_j}$ and $\mathbf{e}_v$ of the target domain. This fine-tuning strategy has a different optimization objective from that of the target domain training. It filters harmful information from source domains [10] while also limiting further improvement of model accuracy.

2) Model Variants: We implement more model variants to compare against alternative designs.

FEDMCDR-*V1: Attention mechanism vs. reinforcement learning.* FEDMCDR uses an RL-based policy agent to filter out negative dimensions. A natural alternative is to use the attention mechanism. Following TrineCDR [12], FEDMCDR-V1 uses SENet to generate attention coefficients for each dimension of the knowledge embeddings.

FEDMCDR-*V2: Rescaling the knowledge.* FEDMCDR uses adaptive mapping functions to align the embedding scales of the source and the target domains. An alternative approach is to directly compute a rescale coefficient c, which leads to variant FEDMCDR-V2: $c = \frac{\|\mathbf{e}^{l}_{u_j}\|_2}{|M| \cdot \|\mathbf{h}^{l}_{S_j}\|_2}$.

FEDMCDR-*V3: Fixed k in the adaptive mapping function.* FEDMCDR has a parameter k in its adaptive mapping function that varies with the training. An alternative strategy is to fix k at a relatively small number throughout the training process. FEDMCDR-V3 takes this approach and sets $k = 1$.

The results are shown in Table VI. FEDMCDR again outperforms all these variants, further validating our model design.

We further plot the values of k for the six different source domains (s1 to s6) in Fig. 5. As it shows, the values of k for

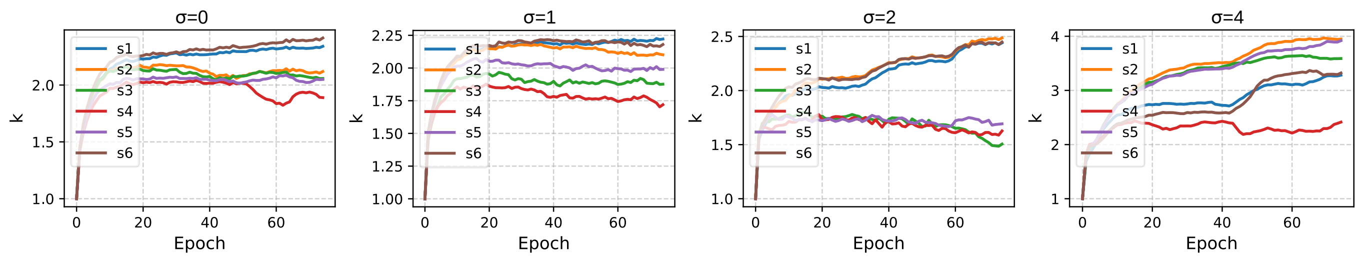

Fig. 5. Evolution of k in FEDMCDR on Amazon-7@Home.

TABLE VII
MODEL ACCURACY ON AMAZON@HOME UNDER MIXED-EVENT SCENARIOS.

Model	H@5	N@5	H@10	N@10
EMCDR	0.1885 ±.0009	0.1343 ±.0006	0.2547 ±.0020	0.1556 ±.0009
PriCDR	0.1897 ±.0030	0.1351 ±.0010	0.2562 ±.0011	0.1565 ±.0004
FedCT	0.1913 ±.0003	0.1370 ±.0001	0.2563 ±.0002	0.1580 ±.0000
FedCDR	0.1584 ±.0051	0.1131 ±.0022	0.2070 ±.0052	0.1288 ±.0019
Fed-CLR	0.0655 ±.0005	0.0384 ±.0003	0.1178 ±.0028	0.0571 ±.0018
FedGCDR	0.2049 ±.0009	0.1434 ±.0003	0.2749 ±.0025	0.1660 ±.0007
FEDMCDR	**0.3400** ±.0072	**0.2449** ±.0052	**0.3828** ±.0091	**0.2588** ±.0059

different domains evolve and diverge with the training process, necessitating adaptive parameters rather than a fixed constant.

F. Robustness against More Complex Negative Transfer (Q4)

We evaluate model robustness against negative knowledge beyond Gaussian noise under two challenging scenarios: a mixed-event scenario featuring negative transfer caused by different types of events and a mixed-noise scenario combining feature dropping with heterogeneous noise distributions.

1) Mixed-event Scenarios: In real-world applications, negative transfer arises from different realistic events, such as domain misalignment, outdated user data, or inaccurate preference signals. To simulate these failures, we partition the target-domain users into four sub-cohorts:

User alignment failure (30% of users): We randomly shuffle the alignments of the same users across different source domains. This subset simulates scenarios where user data from different domains are mismatched due to the absence of a global user identifier across domains. For any user (denoted as u_A) in this 30% subset, its interaction matrix set of source domains are changed from the original set $\{\mathbf{R}^1_A, \mathbf{R}^2_{u_A}, \dots, \mathbf{R}^M_{u_A}\}$ to a set composed of interaction matrices randomly chosen from other users $\{\mathbf{R}^1_{u_D}, \mathbf{R}^2_{u_C}, \dots, \mathbf{R}^M_{u_E}\}$. These misaligned user data creates negative knowledge for the target domain, which would impact the accuracy of existing federated recommender models. For our FEDMCDR, however, this is not an issue because the data storage and knowledge transfer occur on the same individual client devices.

Label inversion (30% of users): For each user, we flip 50% of the observed positive interactions ($r = 1$ in $\mathbf{R}$) to zero and flip an equal number of negative entries ($r = 0$ in $\mathbf{R}$) to one.

Knowledge obsolescence (30% of users): We train the source-domain models using only the first 50% of the interaction timeline, leaving the remaining interaction history unobserved during transfer.

Normal behavior (10% of users): The remaining 10% of users retain unperturbed interactions.

As shown in Table VII, baseline CDR and federated CDR methods suffer significant performance degradation under these structural and semantic perturbations. Traditional mapping functions (e.g., EMCDR [18] and PriCDR [30]) struggle to prevent misleading gradients from corrupting target representations. FEDMCDR achieves substantial superiority, with an HR@10 of 0.3828, compared with 0.2749 of the strongest baseline, FedGCDR. This gain is attributed to our RL-based policy agent and adaptive mapping function, which dynamically isolate fine-grained dimensional conflicts and zero out unhelpful or misleading cross-domain channels.

2) Mixed-noise Scenarios: To further test model robustness against compound noise types, we construct a perturbed embedding $\tilde{\mathbf{h}}$ transmitted from the source domains as:

$$\tilde{\mathbf{h}} = \mathbf{h} \odot \mathbf{m}_{\text{drop}} + \alpha_1 \boldsymbol{\epsilon}_{\text{Gaussian}} + \alpha_2 \boldsymbol{\epsilon}_{\text{Laplacian}} + \alpha_3 \boldsymbol{\epsilon}_{\text{Uniform}}, \quad (40)$$

where $\mathbf{m}_{\text{drop}} \in \{0, 1\}^d$ is a multiplicative drop mask with a feature-missing rate of 0.1. The coefficients $\alpha_1, \alpha_2, \alpha_3 \geq 0$ are randomly sampled for each user under the constraint

$$\alpha_1 + \alpha_2 + \alpha_3 = 1. \quad (41)$$

The variances of the Gaussian, Laplacian, and uniform noise are aligned using a single scale hyperparameter $\sigma \in \{1, 2, 4\}$.

Table VIII presents the results. As σ increases from 1 to 4, while baselines such as PriCDR experience severe accuracy degradation under heavy noise at $\sigma = 4$, FEDMCDR effectively preserves recommendation quality, achieving an HR@10 of 0.2980. These results validate the theoretical bound established in Section V and demonstrate the strong resilience of FEDMCDR to diverse real-world feature distortions.

G. Robustness under Non-zero Domain Bias (Q5)

We conduct mean-shifted noise experiments where the noise mean μ is randomly sampled from three non-zero intervals representing increasing intensities of systematic bias: $|\mu| \in (0, 1]$, $|\mu| \in (1, 2]$, and $|\mu| \in (2, 4]$. Note that we adopt dynamic range intervals rather than fixed values (e.g., $\mu \in \{1, 2, 4\}$) because, without prior knowledge of the target domain's exact embedding mean, a fixed offset like $\mu = 4.0$ does not strictly guarantee a greater distribution shift than $\mu = 2.1$.

TABLE VIII
MODEL ACCURACY ON AMAZON@HOME UNDER MIXED-NOISE SCENARIOS.

Model	$\sigma = 1$				$\sigma = 2$				$\sigma = 4$			
	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10
EMCDR	0.1820 ±.0002	0.1311 ±.0004	0.2356 ±.0004	0.1484 ±.0005	0.1829 ±.0006	0.1313 ±.0003	0.2353 ±.0009	0.1482 ±.0001	0.1827 ±.0009	0.1313 ±.0004	0.2351 ±.0008	0.1482 ±.0002
PriCDR	0.1586 ±.0020	0.1161 ±.0006	0.2108 ±.0016	0.1329 ±.0008	0.1020 ±.0027	0.0726 ±.0018	0.1565 ±.0015	0.0900 ±.0015	0.0662 ±.0027	0.0420 ±.0012	0.1186 ±.0023	0.0587 ±.0010
FedCT	0.1872 ±.0007	0.1334 ±.0005	0.2470 ±.0007	0.1527 ±.0005	0.1872 ±.0007	0.1334 ±.0005	0.2470 ±.0007	0.1527 ±.0005	0.1871 ±.0007	0.1333 ±.0005	0.2471 ±.0007	0.1527 ±.0005
FedCDR	0.1418 ±.0029	0.0924 ±.0026	0.1987 ±.0044	0.0966 ±.0014	0.1374 ±.0019	0.0886 ±.0016	0.1973 ±.0059	0.0934 ±.0018	0.1248 ±.0025	0.0779 ±.0029	0.1893 ±.0025	0.0852 ±.0013
Fed-CLR	0.0650 ±.0007	0.0393 ±.0013	0.1168 ±.0027	0.0560 ±.0011	0.0634 ±.0007	0.0398 ±.0024	0.1181 ±.0030	0.0591 ±.0001	0.0639 ±.0004	0.0410 ±.0015	0.1202 ±.0010	0.0593 ±.0023
FedGCDR	0.2011 ±.0024	0.1421 ±.0009	0.2710 ±.0029	0.1649 ±.0007	0.1987 ±.0015	0.1397 ±.0015	0.2652 ±.0016	0.1614 ±.0011	0.1959 ±.0025	0.1384 ±.0011	0.2622 ±.0002	0.1600 ±.0004
FEDMCDR	**0.3206** ±.0066	**0.2290** ±.0063	**0.3706** ±.0057	**0.2452** ±.0059	**0.2835** ±.0080	**0.2001** ±.0056	**0.3424** ±.0062	**0.2193** ±.0048	**0.2301** ±.0038	**0.1609** ±.0020	**0.2980** ±.0028	**0.1829** ±.0016

TABLE IX
MODEL ACCURACY ON AMAZON@HOME WITH VARYING μ.

Model	$0 < \|\mu\| \leq 1$				$1 < \|\mu\| \leq 2$				$2 < \|\mu\| \leq 4$			
	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10
EMCDR	0.1827 ±.0005	0.1312 ±.0005	0.2356 ±.0004	0.1483 ±.0003	0.1831 ±.0009	0.1316 ±.0005	0.2356 ±.0006	0.1485 ±.0004	0.1830 ±.0009	0.1315 ±.0006	0.2358 ±.0009	0.1485 ±.0004
PriCDR	0.0538 ±.0010	0.0312 ±.0006	0.1053 ±.0016	0.0478 ±.0008	0.0493 ±.0019	0.0291 ±.0012	0.0972 ±.0007	0.0446 ±.0005	0.0432 ±.0018	0.0256 ±.0009	0.0856 ±.0021	0.0397 ±.0009
FedCT	0.1864 ±.0008	0.1336 ±.0005	0.2456 ±.0007	0.1527 ±.0007	0.1864 ±.0007	0.1325 ±.0007	0.2453 ±.0017	0.1515 ±.0007	0.1860 ±.0012	0.1323 ±.0007	0.2422 ±.0013	0.1505 ±.0005
FedCDR	0.0904 ±.0020	0.0546 ±.0019	0.1555 ±.0020	0.0674 ±.0016	0.0891 ±.0029	0.0537 ±.0022	0.1568 ±.0017	0.0663 ±.0011	0.0835 ±.0073	0.0493 ±.0049	0.1534 ±.0011	0.0641 ±.0025
Fed-CLR	0.0646 ±.0010	0.0386 ±.0009	0.1188 ±.0025	0.0581 ±.0016	0.0658 ±.0002	0.0390 ±.0021	0.1177 ±.0018	0.0566 ±.0002	0.0628 ±.0017	0.0388 ±.0017	0.1189 ±.0019	0.0587 ±.0012
FedGCDR	0.1946 ±.0019	0.1378 ±.0005	0.2612 ±.0018	0.1594 ±.0006	0.1957 ±.0022	0.1377 ±.0011	0.2606 ±.0010	0.1586 ±.0007	0.1955 ±.0019	0.1365 ±.0010	0.2611 ±.0023	0.1561 ±.0013
FEDMCDR	**0.2053** ±.0029	**0.1444** ±.0016	**0.2737** ±.0030	**0.1665** ±.0015	**0.2007** ±.0020	**0.1421** ±.0014	**0.2692** ±.0029	**0.1641** ±.0018	**0.1957** ±.0018	**0.1372** ±.0014	**0.2657** ±.0013	**0.1589** ±.0012

TABLE X
MODEL ACCURACY ON AMAZON@HOME WITH PARTIAL USER SET OVERLAPS WHEN $\sigma = 0$.

Model	H@5	N@5	H@10	N@10
EMCDR	0.1884 ±.0009	0.1342 ±.0004	0.2562 ±.0006	0.1561 ±.0004
PriCDR	0.1885 ±.0020	0.1346 ±.0010	0.2581 ±.0018	0.1571 ±.0004
FedCT	0.1911 ±.0007	0.1357 ±.0005	0.2589 ±.0005	0.1576 ±.0004
FedCDR	0.1637 ±.0025	0.1172 ±.0014	0.2114 ±.0044	0.1326 ±.0019
Fed-CLR	0.0645 ±.0005	0.0417 ±.0025	0.1203 ±.0026	0.0564 ±.0021
FedGCDR	0.2035 ±.0018	0.1432 ±.0005	0.2726 ±.0015	0.1657 ±.0005
FEDMCDR	**0.3574** ±.0037	**0.2586** ±.0031	**0.3960** ±.0057	**0.2711** ±.0038

Table IX reports the accuracy on Amazon@Home under these varied mean offsets. As expected, models relying on static feature mapping or rigid attention schemes (e.g., PriCDR and FedCDR) experience significant performance declines as the magnitude of systematic offset $|\mu|$ scales up. FEDMCDR again consistently achieves higher accuracy across all bias intervals. This robustness is attributed to two key designs in FEDMCDR: (1) the adaptive mapping function $f_A^i(\cdot)$, which rescales and dynamically normalizes transferred representations into the target domain's latent space, thereby absorbing constant systematic domain shifts; and (2) the pre-trained RL-based policy agent, which performs dimension-level pruning to mask out features corrupted by severe feature conflicts or spurious domain correlations before graph expansion.

H. Evaluation under Partial User Set Overlaps (Q6)

To evaluate FEDMCDR under partial user set overlap scenarios, we randomly mask cross-domain user associations.

For each target domain user $u \in U$ and each source domain S_i, we introduce an independent Bernoulli sampling mask $b_{u,i} \sim \text{Bernoulli}(0.5)$. When $b_{u,i} = 1$, the cross-domain link is retained, allowing source domain S_i to transfer knowledge for user u. Otherwise, the cross-domain link is removed, simulating scenario where no virtual user node corresponding to u of S_i is constructed during target domain graph expansion. This setup reduces the average user set overlap rate between the target domain and any given source domain to 50%.

Tables X and XI report the recommendation accuracy on Amazon@Home with partial user set overlaps for $\sigma = 0$ and $\sigma \in \{1, 2, 4\}$, respectively. When 50% of user cross-domain associations are unavailable, FEDMCDR significantly outperforms all competing models. For example, at $\sigma = 0$, FEDMCDR achieves an HR@10 of 0.3960, achieving a relative improvement of 45.3% over the strongest baseline FedGCDR (0.2726).

VIII. CONCLUSION

We proposed FEDMCDR, a federated model that generalizes cross-domain recommendation to multiple source domains. To mitigate negative transfer, we proposed a reinforcement learning-based dimension filter technique over user preference embeddings and an adaptive graph expansion algorithm that propagates only the most relevant cross-domain knowledge. Through theoretical analysis, we show

TABLE XI
MODEL ACCURACY ON AMAZON@HOME WITH PARTIAL USER SET OVERLAPS.

Model	$\sigma = 1$				$\sigma = 2$				$\sigma = 4$			
	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10	H@5	N@5	H@10	N@10
EMCDR	0.1891 ±.0005	0.1349 ±.0001	0.2566 ±.0007	0.1566 ±.0003	0.1885 ±.0009	0.1343 ±.0006	0.2570 ±.0010	0.1564 ±.0002	0.1888 ±.0009	0.1346 ±.0006	0.2565 ±.0007	0.1565 ±.0003
PriCDR	0.1359 ±.0031	0.0994 ±.0014	0.1861 ±.0026	0.1154 ±.0013	0.0820 ±.0027	0.0552 ±.0012	0.1344 ±.0028	0.0719 ±.0012	0.0545 ±.0013	0.0323 ±.0011	0.1067 ±.0018	0.0489 ±.0006
FedCT	0.1916 ±.0007	0.1360 ±.0002	0.2586 ±.0004	0.1577 ±.0001	0.1913 ±.0006	0.1361 ±.0002	0.2589 ±.0005	0.1579 ±.0003	0.1911 ±.0009	0.1360 ±.0004	0.2593 ±.0009	0.1580 ±.0002
FedCDR	0.1244 ±.0032	0.0866 ±.0013	0.1632 ±.0051	0.0961 ±.0007	0.1169 ±.0066	0.0797 ±.0047	0.1572 ±.0084	0.0863 ±.0064	0.0945 ±.0021	0.0572 ±.0020	0.1503 ±.0024	0.0688 ±.0009
Fed-CLR	0.0627 ±.0016	0.0416 ±.0005	0.1168 ±.0012	0.0595 ±.0003	0.0655 ±.0021	0.0384 ±.0019	0.1204 ±.0023	0.0596 ±.0019	0.0655 ±.0015	0.0394 ±.0017	0.1172 ±.0031	0.0582 ±.0003
FedGCDR	<u>0.1997</u> ±.0024	<u>0.1402</u> ±.0008	<u>0.2674</u> ±.0023	<u>0.1621</u> ±.0005	<u>0.1962</u> ±.0007	<u>0.1382</u> ±.0008	<u>0.2625</u> ±.0015	<u>0.1597</u> ±.0004	<u>0.1948</u> ±.0018	<u>0.1376</u> ±.0007	<u>0.2625</u> ±.0015	<u>0.1596</u> ±.0003
FEDMCDR	**0.3227** ±.0058	**0.2303** ±.0044	**0.3703** ±.0052	**0.2458** ±.0040	**0.2620** ±.0090	**0.1850** ±.0071	**0.3217** ±.0078	**0.2044** ±.0066	**0.2136** ±.0028	**0.1494** ±.0022	**0.2803** ±.0042	**0.1710** ±.0027

that FEDMCDR bounds the impact of negative knowledge in model training for the target domain. Extensive experiments on Amazon and Douban datasets demonstrate that FEDMCDR consistently achieves strong accuracy. Compared with the strongest baseline, FEDMCDR yields relative improvement of up to 42.0% in HR@10 and 57.5% in NDCG@10.

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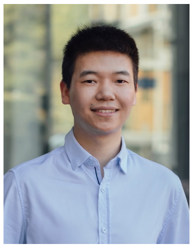

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